Cobweb Forests Are Universal Approximators

Draft — working notes

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Abstract

Cobweb learns a hierarchy of probabilistic concepts one example at a time, without gradients, by choosing among four operators with category utility. We ask whether Cobweb can represent and learn *any* continuous function, and answer the question for single trees and for *Cobweb forests*, networks in which the predictions of some trees are the inputs of others. (1) One Cobweb tree is a universal approximator. (2) Forests in which every tree reads at most two inputs are universal approximators, and one input per tree is not enough. (3) Forests exploit composition: if the target is a composition of k-input maps, a forest whose trees have G leaves each approximates it with error of order $G^{-1/k}$, whatever the number n of inputs, while the bound for one tree is $G^{-1/n}$. (4) Cobweb never deletes what it stores, so the tree it learns is a consistent regressor whatever hierarchy category utility builds, and a learned forest converges to the forest of conditional expectations its trees are asked to predict; we characterise which of these assignments recover the target. Every theorem predicts a measurable quantity—a rate, a floor, an identity—and we measure each one on real Cobweb trees; the measurements agree with the predictions, the one exception being a single tree in eight dimensions, which is still pre-asymptotic at the largest grid we can build. We close with a recipe for building a forest from data.

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1 Introduction

A model class is a *universal approximator* if, for every continuous function on a compact set and every tolerance, some member of the class is uniformly within that tolerance. Neural networks with one hidden layer are universal [4, 12], and the Kolmogorov–Arnold theorem [1, 13], which writes every continuous function of n variables through one-variable functions and addition, has recently inspired an architecture with learnable one-variable units [14]. Universality says what a model *can* represent; it is the first thing one wants to know about a new model class.

Cobweb [10] is a model class of a different kind. It sorts each example down a tree of probabilistic concepts and, at every node, takes whichever of four operators (add to the best child, create a child, merge two children, split a child) most increases *category utility*, the information the concepts carry about the features [3, 17]. It learns incrementally, in closed form, and without gradients. Several systems already wire Cobweb trees together: Convolutional Cobweb [16] feeds the concept labels of patch trees to another tree, TRELLIS [20] describes composites by concepts of another tree, and a Cobweb language model passes each tree's prediction to the trees above it. We call such a network a *Cobweb forest*. This paper asks:

Q1. Is a Cobweb tree a universal approximator? Is a Cobweb forest, and with how many inputs per tree?

Q2. What does composing trees buy?
Q3. Does Cobweb's own learning procedure learn the function, for a tree and for a forest?

Answers.

- One tree is universal (Theorem 1): a tree whose leaves hold the target at the points of a fine net, predicting with a small variance floor, is uniformly accurate.
- Forests of two-input trees are universal (Theorem 4); forests of one-input trees are not (Proposition 5). The proof composes trees that add and multiply, and uses the Stone–Weierstrass theorem; the Kolmogorov–Arnold theorem gives a second construction with exactly $4n^2 + 4n$ trees.
- Composition beats dimension (Theorem 7): for targets composed of k-input maps a forest attains error $O(G^{-1/k})$ with G leaves per tree, against the single-tree bound $O(G^{-1/n})$. This is the sense in which tree–tree interactions are worth having.
- Cobweb's learning is consistent (Theorem 8), and a learned forest converges to an explicit limit (Theorem 9). Whether the limit is the target depends on what each tree is asked to predict (Proposition 10), and no tree can tell from its own data what it should predict (Proposition 11).

Each answer comes with a measurement on real Cobweb trees (Section 5, Table 1): the predicted rates, floors and identities are the measured ones. Section 6 turns the theory into a recipe for building a forest.

2 Cobweb trees and forests

A tree as a function. A Cobweb tree stores examples (x, y), $x \in \mathbb{R}^d$. Every concept c keeps the number n_c of examples below it and their mean $(\mu_{c,x}, \mu_{c,y})$ and variance; with exact matching, a leaf holds n_c copies of one example, so its spread is zero and its Gaussian has the variance floor h^2 alone. To predict y at x, the tree mixes the means of its leaves, weighting each by its count and its likelihood of x:

$$\hat{y}(x) = \frac{\sum_{c \text{ leaf}} n_c \, e^{-\|x-\mu_{c,x}\|^2/2h^2} \, \mu_{c,y}}{\sum_{c \text{ leaf}} n_c \, e^{-\|x-\mu_{c,x}\|^2/2h^2}}. \tag{1}$$

(The Gaussian normalising constants are equal for all leaves and cancel.) This *count-weighted leaf mixture* is one of the prediction rules of the implementation; Appendix A defines all of them and says which results cover which rule.

Forests. A *Cobweb forest* is a directed acyclic graph of Cobweb trees with one output tree. Each tree reads some raw coordinates of x and the *messages* of its parents: their *activations*, the predictions (1) of their outcomes, or their *labels*, the concepts the input sorts to. The forest's output is the output tree's activation. The *fan-in* of a tree is the number of its inputs. Every tree learns with Cobweb's own procedure on examples made of its inputs and its *outcome*, the variable it is asked to predict (Definitions A.5 and D.1).

3 Representation

3.1 One tree

Theorem 1 (One Cobweb tree is a universal approximator)**.** *Let $K \subset \mathbb{R}^d$ be compact, $g : K \to \mathbb{R}$ continuous and $\varepsilon > 0$. There are a finite set $S \subseteq K$ and $h > 0$ such that the depth-one Cobweb tree whose leaves hold the examples $(s, g(s))$, $s \in S$, satisfies $|\hat{y}(x) - g(x)| < \varepsilon$ for every $x \in K$.*

Proof. Let $M = \max_K |g|$ (finite, since K is compact). Since g is uniformly continuous on K, there is $\rho > 0$ with $|g(x) - g(z)| < \varepsilon/2$ whenever $x, z \in K$ and $\|x - z\| < \rho$. Since K is compact, it has a finite $\rho/2$-net: a finite $S \subseteq K$ such that every $x \in K$ is within $\rho/2$ of some $s \in S$. Give every leaf count one. By (1), $\hat{y}(x) = \sum_{s \in S} w_s(x) g(s)$ with $w_s(x) = a_s(x) / \sum_{t \in S} a_t(x)$ and $a_s(x) = e^{-\|x-s\|^2/2h^2}$.

Fix $x \in K$ and $s_0 \in S$ with $\|x - s_0\| < \rho/2$, so $\sum_t a_t(x) \geq a_{s_0}(x) > e^{-\rho^2/8h^2}$. Call s *far* if $\|x - s\| \geq \rho$; then $a_s(x) \leq e^{-\rho^2/2h^2}$, so the far leaves have total weight

$$\sum_{s \text{ far}} w_s(x) \;\leq\; |S|\, e^{-\rho^2/2h^2}\, e^{\rho^2/8h^2} = |S|\, e^{-3\rho^2/8h^2}.$$

Since $\sum_s w_s(x) = 1$,

$$|\hat{y}(x) - g(x)| \leq \sum_{s \text{ near}} w_s(x)\, |g(s) - g(x)| + \sum_{s \text{ far}} w_s(x)\, |g(s) - g(x)| < \frac{\varepsilon}{2} + 2M|S|\, e^{-3\rho^2/8h^2},$$

where the near leaves contribute less than $\varepsilon/2$ because each is within ρ of x. Choose $h > 0$ so small that $2M|S|e^{-3\rho^2/8h^2} \leq \varepsilon/2$; the bound is then below ε for every $x \in K$. □

The proof uses only that K is compact and g continuous; it works verbatim for any positive counts, with $|S|$ replaced by $|S| \max_s n_s / \min_s n_s$. For a Lipschitz g a regular grid gives an explicit rate:

Proposition 2 (Rate; Theorem B.2)**.** *If g is L-Lipschitz and K lies in a cube of side w, a depth-one tree with $G = m^d$ leaves on a grid, with h proportional to the grid spacing, has $\sup_K |\hat{y} - g| \leq C_d\, L\, w\, G^{-1/d}$.*

The implementation's other prediction rules mix over the root as well as the leaves. Two of them (LIK, PATH) are universal by the same construction, because the root's weight is of order $1/G$; the other two (PRIOR, POSTERIOR) keep a fixed share on the root of a flat tree and are not (Theorem B.2 and Proposition B.3).

3.2 Forests

Lemma 3 (Stability of composition; Lemma B.5)**.** *Let a finite DAG of continuous maps g_v compute F on a compact K. For every $\varepsilon > 0$ there are $\rho_v, \eta_v > 0$ such that replacing every g_v by any map $\hat{g}_v$ with $|\hat{g}_v - g_v| \leq \eta_v$ on the ρ_v-neighbourhood of g_v's input set changes the output by less than ε on K.*

Theorem 4 (Forests of two-input trees are universal approximators)**.** *Let $K \subset \mathbb{R}^n$ be compact, $f : K \to \mathbb{R}$ continuous and $\varepsilon > 0$. There is a Cobweb forest in which every tree has at most two inputs and passes its activation, whose output is within ε of f on K.*

Proof. By the Stone–Weierstrass theorem [21] the polynomials in $x_1, \ldots, x_n$ are dense in $C(K)$: they form an algebra of continuous functions that contains the constants and separates points. Take a polynomial p with $\max_K |p - f| < \varepsilon/2$. Every polynomial is computed by a finite DAG whose sources are the coordinates x_i and constants and whose other vertices apply $(a, b) \mapsto a + b$ or $(a, b) \mapsto ab$ (by induction on the expression). By Lemma 3 with tolerance $\varepsilon/2$ there are ρ_v, η_v for this DAG. Each vertex map is continuous on the compact set $K_v^{\rho_v}$, the closed ρ_v-neighbourhood of its input set, so by Theorem 1 a Cobweb tree with two inputs approximates it within η_v there; a constant is a tree with no input. The forest of these trees is within $\varepsilon/2$ of p, hence within ε of f, on K. □

Proposition 5 (One input per tree is not enough; Proposition B.7)**.** *If every tree of a forest has at most one input, the output is a function of a single coordinate.*

Remark 6 (An explicit size: the Kolmogorov–Arnold forest; Theorem B.6)*.* The Kolmogorov–Arnold representation

$$f(x) = \sum_{q=0}^{2n} \Phi_q\Big(\sum_{p=1}^{n} \varphi_{q,p}(x_p)\Big) \tag{2}$$

is a DAG of $n(2n+1)$ inner maps, $(2n+1)(n-1)$ additions, $2n+1$ outer maps and $2n$ additions. The same argument turns it into a forest of $4n^2+4n$ trees with at most two inputs each, a size that does not depend on f or ε (only the trees' leaves do).

Two further facts place these results. Trees that pass *labels* are also universal: trees that categorize each coordinate, feeding a lookup tree, approximate any f, but the lookup tree needs one leaf per cell of the product partition (Theorem B.9). And merging trees under one common root does not compose them: the merged tree predicts a convex combination of their predictions (Proposition B.11). Composition is what a forest adds to a tree.

3.3 What composition buys

Universality says nothing about size. Many functions of interest are compositions of functions of few variables, and for them a forest escapes the dimension. Call f a *k-composition of depth D* if it is computed by a DAG whose sources are the coordinates and whose every other vertex v applies a map g_v of at most k of its predecessors, the longest path having D such vertices.

Theorem 7 (Composition beats dimension)**.** *Let f be a k-composition of depth D on a compact $K \subset \mathbb{R}^n$. Let B_v be the set of input vectors of g_v as x ranges over K, and suppose that, for some $r > 0$, every g_v is L-Lipschitz in the max-norm on the r-neighbourhood B_v^r of B_v, which lies in a cube of side w. For $G = m^k$, $m \in \mathcal{N}$, let $\eta = C_k L w G^{-1/k}$ be the bound of Proposition 2. If $\eta \sum_{l=0}^{D-1} L^l \le r$, the forest with one tree of G leaves per vertex, each built by Proposition 2 on B_v^r, satisfies*

$$\sup_K |\hat{F} - f| \;\le\; C_k\, L\, w\Big(\sum_{l=0}^{D-1} L^l\Big) G^{-1/k},$$

a rate set by the fan-in k, not by n. The corresponding bound for one tree over all n coordinates is $C_n\, L_f \operatorname{diam}(K)\, G^{-1/n}$ (Proposition 2).

Proof. Write $e_v = \sup_K |\hat{u}_v - u_v|$ for the error of the value at vertex v ($e = 0$ at the sources). Suppose every predecessor of v has error at most $e \le r$. Then v's approximate input vector is within e (in max-norm) of its exact input vector, so it lies in B_v^r, where the tree is within η of g_v and g_v is L-Lipschitz:

$$|\hat{g}_v(\hat{u}) - g_v(u)| \le |\hat{g}_v(\hat{u}) - g_v(\hat{u})| + |g_v(\hat{u}) - g_v(u)| \le \eta + Le.$$

By induction on the depth l of a vertex, $e_v \le \eta \sum_{j<l} L^j$; the hypothesis keeps every $e_v \le r$. At the output, $l \le D$. □

The bound for one tree is only an upper bound, so the theorem does not prove that a single tree *must* do worse; the measurements of Section 5 show that it does, with the predicted exponents. The same dichotomy between compositional and generic targets is known for deep and shallow networks [19].

4 Learning

The constructions above place leaves by hand. We now let Cobweb learn.

4.1 One tree

Theorem 8 (Cobweb's learning is consistent; Lemma C.1 and Theorem C.2)**.** *None of Cobweb's operators deletes a leaf, and with exact matching (and below the depth limit) every example ends in a leaf that holds only copies of it. Hence, whatever hierarchy category utility builds and in whatever order the examples arrive, the count-weighted leaf mixture* (1) *of the learned tree* equals *the Nadaraya–Watson kernel regression on all examples seen. Consequently, if* $h \to 0$ *and* $Nh^d \to \infty$*, the learned tree converges to* $\mathbb{E}[Y \mid X]$ *in* L_1*, almost surely, for every distribution with bounded* Y *[8]; and if* $\mathbb{E}[Y \mid X = x]$ *is twice differentiable and* X *has a smooth density bounded below, then with* $h \propto N^{-1/(d+4)}$ *its root-mean-squared error on any compact set interior to the support is* $O(N^{-2/(d+4)})$ *[24].*

The hierarchy plays no role in consistency; it matters for search under a budget, for the default PATH rule, and for the labels a tree passes on (Remark C.4). The same fact—nothing stored is deleted—means a tree does not forget: learning new examples changes old predictions only through the kernel tails of the new leaves.

4.2 Forests

A tree in a forest learns to predict its *outcome* V_τ from its inputs U_τ. By Theorem 8 it learns $\mathbb{E}[V_\tau \mid U_\tau]$, so a forest has a natural limit, its *ideal forest*: in topological order every tree emits $m_\tau = \mathbb{E}[V_\tau \mid U_\tau]$, with U_τ built from its parents' ideal messages; the ideal output is F. The *deficit* $\Delta = \mathbb{E}(f - F)^2$, with $f = \mathbb{E}[Y \mid X]$, measures what the assignment of outcomes loses (Definition D.3).

Theorem 9 (A learned forest converges to its ideal forest; Theorem D.5 and Corollary D.6)**.** *Consider a forest whose trees pass activations, trained level by level on separate parts of the data (the split protocol). Under regularity conditions (Assumption D.4) its output converges uniformly, in probability, to* F*, and its mean squared error to* $\Delta + \mathbb{E}\operatorname{Var}(Y \mid X)$*. It is consistent if and only if* $\Delta = 0$*.*

A KAN or a neural network learns inner functions from the error of their composition; a Cobweb tree receives no error and can only learn to predict something it will observe. The natural observable outcomes are the target Y (*funnel*), a part of the input the tree does not see (*cross-view*), or none, in which case the tree passes labels (*categorization*).

Proposition 10 (Which outcomes recover the target; Appendix D)**.**

(a) If $Y = \Phi\big(\sum_j g_j(X_{B_j})\big) + \xi$ *with* Φ *continuous and strictly increasing, continuous* g_j*, independent blocks* X_{B_j} *and independent noise, every funnel forest whose graph is a rooted tree reading each block in exactly one tree has* $\Delta = 0$ *(Proposition D.7).*

(b) If $Y = X_1 X_2$ *with independent inputs symmetric about* 0*, the funnel forest over two one-input trees has* $F \equiv 0$ *and* $\Delta = \operatorname{Var} Y$*: it learns nothing (Proposition D.8).*

(c) Label forests trained with the split protocol are consistent for every target whenever their cells shrink, stay few compared with the data, and their smoothing vanishes fast enough (Theorem D.10).

result	quantity	predicted	measured
Proposition 2	one constructed tree, $d = 1$: rate in G	-1.000	-1.058
	one constructed tree, $d = 2$: rate in G	-0.500	-0.511
	one constructed tree, $d = 3$: rate in G	-0.333	-0.298
Theorem 7	Lipschitz maps, $n = 2$: forest / one tree, rate in G	-0.500 / -0.500	-0.603 / -0.541
	Lipschitz maps, $n = 4$: forest / one tree, rate in G	-0.500 / -0.250	-0.400 / -0.229
	Lipschitz maps, $n = 8$: forest / one tree, rate in G	-0.500 / -0.125	-0.487 / -0.056
	smooth maps, $n = 2$: forest / one tree, rate in G	≤ -0.500 / ≤ -0.500	-0.915 / -0.608
	smooth maps, $n = 4$: forest / one tree, rate in G	≤ -0.500 / ≤ -0.250	-0.825 / -0.282
	smooth maps, $n = 8$: forest / one tree, rate in G	≤ -0.500 / ≤ -0.125	-0.739 / -0.107
Theorem 8	learned tree = Nadaraya–Watson: identity	0	$2.9e-07$
	one learned tree, $d = 1$: interior rate in N	-0.400	-0.413
	one learned tree, $d = 2$: interior rate in N	-0.333	-0.334
	one learned tree, $d = 3$: interior rate in N	-0.286	-0.307
Proposition B.11	merged trees = mixture: identity	0	$1.9e-06$
Proposition B.3	PRIOR rule, flat tree: error floor	> 0	0.50
Proposition 10(b)	funnel forest on x_1x_2: floor ($N = 16000$)	1	1.000
Proposition 10(c)	label forest on x_1x_2: floor ($N = 16000$)	$\to 0$	0.083
Theorem 8	one tree on x_1x_2 ($N = 16000$)	$\to 0$	0.023
Proposition 10(d)	cross-view forest, view-specific target: floor ($N = 16000$)	0.696	0.716
Remark D.13	learned forest / one tree, $n = 4$: rate in N (heuristic)	-0.333 / -0.250	-0.357 / -0.324
	learned forest / one tree, $n = 8$: rate in N (heuristic)	-0.333 / -0.167	-0.383 / -0.118
Proposition 11	law of (x_2, Y) in the two worlds	equal	equal (exact; Lean)

Table 1: Each prediction of the theory and its measurement. "rate": exponent of the error against the number of leaves G or of examples N. "floor": limit of the error divided by $\mathrm{sd}(f)$. "identity": largest deviation from an identity that holds exactly.

(d) In a Gaussian two-view model, cross-view forests have $\Delta = 0$ when the target depends on what the views share, and an explicitly computable $\Delta > 0$ when it does not (Proposition D.9).

Proposition 11 (No tree can tell its role from its own data; Theorem E.2 and Corollary E.3)**.** *Let X_1, X_2, X_3 be independent and uniform on $[-1, 1]$, and observe (X_1, X_2, Y) with either $Y = \mathrm{sign}(X_1X_2)$ or $Y = \mathrm{sign}(X_1X_3)$. In the first world X_2 is essential, in the second irrelevant, yet (X_2, Y) has the same distribution in both; any rule that decides from the data of a tree reading X_2 whether to keep it errs with probability at least $1/2$ in one world.*

What a tree should predict must therefore be decided where trees are composed: choosing among candidate forests by the validation error of the output is consistent (Theorem E.4).

5 Theory meets measurement

Every result above predicts a number. Table 1 lists each prediction next to its measurement on real Cobweb trees—constructed exactly where the result is about existence, learned with Cobweb's own procedure where it is about learning. Learned trees predict with the count-weighted leaf mixture under full expansion and forests use the split protocol, the setting of the theorems; the variance floor follows the schedule of Theorem 8 with its constant chosen on validation data. Rates are least-squares slopes in log–log coordinates over the sizes shown in the figures; errors are averaged over three seeds. Appendix F gives the details.

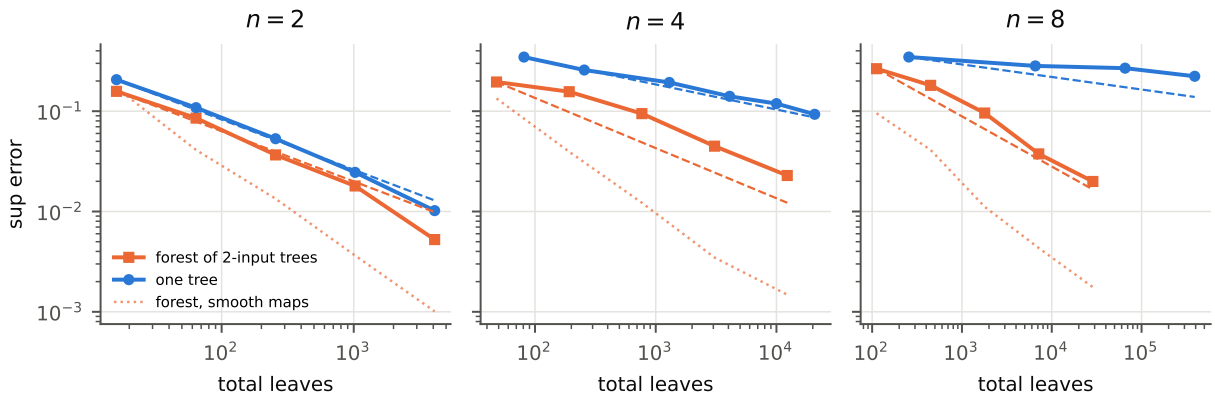


Figure 1: Composition beats dimension (Theorem 7). Sup error of constructed Cobweb trees and forests on a composition of two-input maps in $n = 2, 4, 8$ dimensions, against the total number of leaves. Dashed: the predicted slopes, $-1/2$ for the forest and $-1/n$ for one tree.

Representation. One constructed tree's sup error falls with the predicted exponent $-1/d$ (measured -1.06, -0.51, -0.30 for $d = 1, 2, 3$). On the composition of two-input maps (Fig. 1) the forest's exponent stays near $-1/2$ in every dimension (-0.60, -0.40, -0.49 for $n = 2, 4, 8$), while the single tree's exponent shrinks with n as predicted (-0.54 and -0.23 against -0.5 and -0.25; at $n = 8$ its grids, with at most five points per axis, are still pre-asymptotic: -0.06 against -0.125). The difference in size is large: in eight dimensions a forest with 28,672 leaves in all has sup error 0.020, a single tree with 390,625 leaves 0.22. With smooth maps both converge faster than the Lipschitz bound (-0.74 to -0.92 for the forest), as an upper bound allows.

Learning. A learned tree's count-weighted leaf mixture equals the Nadaraya–Watson estimate to within 3×10^{-7}, float precision, and its interior error falls at -0.41, -0.33, -0.31 for $d = 1, 2, 3$ against the predicted -0.40, -0.33, -0.29. For learned forests we have a limit (Theorem 9) but no proved rate. A heuristic (Remark D.13) says each tree regresses on its own $k = 2$ inputs, at $N^{-2/(k+4)} = N^{-1/3}$; the forests follow it (-0.36 for $n = 4$, -0.38 for $n = 8$). One tree over all inputs converges at least as fast as its exponent $-2/(n + 4)$ for $n = 4$ (-0.32) and is still pre-asymptotic for $n = 8$ (-0.12). In eight dimensions at $N = 16{,}000$ the forest's normalised error is 0.25 and the single tree's 0.74 (Fig. 2).

What trees predict. Each kind of outcome behaves as Proposition 10 says. On x_1x_2 the funnel forest's normalised error is 1.00 at every N, the predicted floor: its trees' activations are constant. The label forest, with its smoothing inside condition (c) of Theorem D.10, falls from 0.24 to 0.13 to 0.083 as N grows from 10^3 to 1.6×10^4, and the validated label depth grows with N (3, 3, 4): its cells refine, as the theorem requires. (With the default smoothing $\alpha = 0.1$, outside condition (c), it stalls near 0.6.) On the view-specific Gaussian target the cross-view forest approaches its predicted floor 0.696 from above (0.81, 0.78, 0.72).

6 A recipe for building a forest

The theory says what a good forest looks like; turning it into a procedure needs only quantities that Cobweb trees can estimate. The recipe below plays the role of the KAN paper's simplification

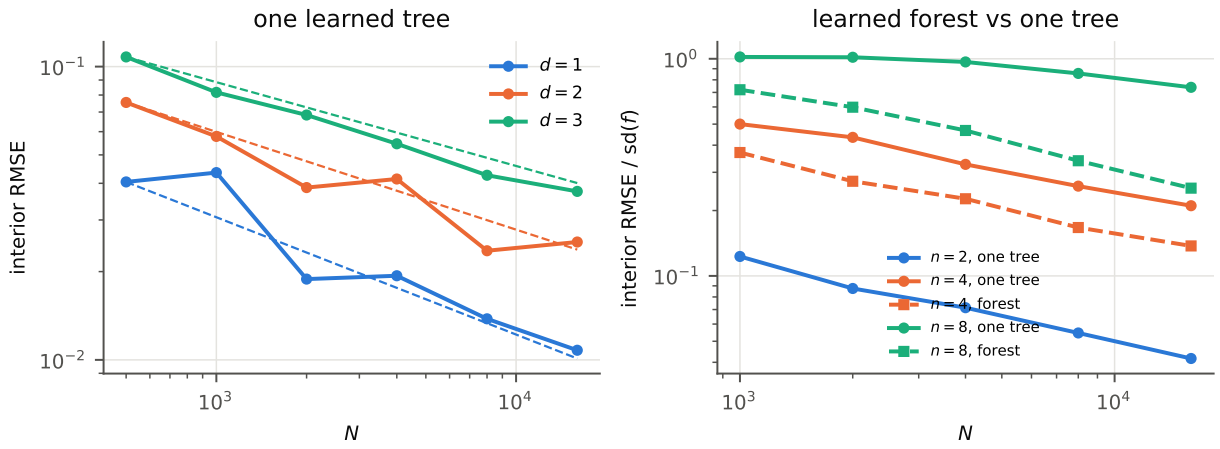


Figure 2: Learning. Left: one learned tree, interior RMSE against N in $d = 1, 2, 3$ dimensions; dashed: the predicted slopes $-2/(d+4)$. Right: learned funnel forest and one learned tree on $\Phi(\sum_j g(X_{B_j}))$ with two-dimensional blocks, $n = 2, 4, 8$.

procedure [14]. Each step is justified by a result above.

Procedure 12 (Building a Cobweb forest)**.**

1. *Start small.* One tree per input coordinate (or per given group), composed in a balanced binary graph of two-input trees (Theorem 4: two inputs suffice).

2. *Merge what interacts.* For every pair of inputs, compare a tree over the pair with two one-input trees composed by activations; merge the pair into one tree when the merged tree is better by more than a threshold (the *synergy deficit*, Definition E.5). Activations lose exactly these interactions (Proposition 10(b)), and merging keeps the fan-in—and so the rate of Theorem 7—small only when few pairs interact.

3. *Give every tree an observable outcome.* The target where the parts combine monotonically (Proposition 10(a)); labels or a merged tree where they interact; another view where the target lives in what views share (Proposition 10(d)).

4. *Decide at the composing tree.* Compare candidate forests—always including the single merged tree—by the output's validation error (Proposition 11, Theorem E.4).

5. *Sharpen with data.* Shrink each tree's variance floor as $h \propto N^{-1/(k+4)}$ with k its fan-in (Theorem 8); this is the analogue of a KAN's grid extension.

6. *Prune and read off.* Drop a tree whose removal does not raise the validation error, and plot every remaining tree's activation against its inputs: on targets of Proposition 10(a) these are the inner functions up to monotone maps.

The recipe is greedy and its threshold is a choice. Pairwise deficits miss interactions of higher order—for the parity of three signs every pair is independent of the target (Corollary E.3)—which is why step 4 always keeps the merged tree. On the Feynman formulas and the four-dimensional block target (Table 6) steps 1–4 recover the groups of interacting variables—$\{B, v, \theta\}$ in $q(E_f + Bv\sin\theta)$, the two angles of $I.29.16$, the two blocks of the 4-D target—and the selected forest is within one

percent of the best candidate in every case but one; where no pair interacts, the recipe returns the single tree.

7 Conclusion

Cobweb is a universal approximator: one tree suffices to represent any continuous function, forests of two-input trees suffice as well, and forests represent compositional functions at a rate set by their fan-in rather than by the dimension. Cobweb's own learning is consistent for a tree, and a learned forest converges to an explicit limit whose distance from the target is decided by what each tree predicts. Every one of these statements predicts a number that the implementation reproduces.

Limitations. The convergence theorems assume the count-weighted leaf mixture, exact matching and, for forests, the split protocol; Theorem D.5 assumes a uniform law for kernel smoothers up to the boundary ((A5) of Assumption D.4). The single-tree rates are upper bounds, and the learned-forest rates of Fig. 2 are measured, not proved. The label conditions of Theorem D.10 are not derived from category utility, and the recipe is greedy.

A cognitive note. The three outcomes a tree can be given are three things a learner can do with what it perceives: categorize it, predict an outcome from it, or predict another part of it. The results say when each suffices, and Proposition 11 is a precise form of the credit-assignment problem [18]: a module cannot learn its role from what it alone sees.

A The model in full

A.1 Concepts and trees

An *instance* is a pair $z = (a, u)$ of a nominal part a (a table that assigns to each attribute a bag of values with weights) and a continuous part $u \in \mathbb{R}^m$.

Definition A.1 (Concept)**.** A concept c stores its count n_c, the attribute–value counts $n_c(A{=}v)$ of the instances it covers, and, for the continuous block, their mean $\mu_c \in \mathbb{R}^m$ and empirical (per-dimension) variance $s_c^2 \in \mathbb{R}^m_{\geq 0}$. With smoothing parameters $\alpha > 0$ and $h^2 > 0$ (`alpha`, `prior_var`) a concept scores a query with

$$P_c(A{=}v) = \frac{n_c(A{=}v) + \alpha}{n_c(A) + (K_A + 1)\alpha}, \qquad p_c(u_j) = \mathcal{N}\big(u_j;\, \mu_{c,j},\, v_{c,j}\big), \quad v_{c,j} = s_{c,j}^2 + h^2,$$

with attributes conditionally independent given c and K_A the number of values of A seen so far (the extra α reserves mass for an unseen value). When a concept *predicts* a missing attribute it uses $(n_c(A{=}v) + \alpha)/(n_c(A) + K_A\alpha)$, which sums to one over the seen values.

Definition A.2 (Cobweb tree)**.** A Cobweb tree $\mathcal{T}$ is a rooted tree of concepts whose statistics are *additive*: each internal concept's statistics are those of the pooled instances of its children. Its *category utility* at a node with children C is the normalised information gain $\mathrm{PU} = |C|^{-1}\big[H(\text{parent}) - \sum_{c \in C} P(c)H(c)\big]$, where H is the sum of the Dirichlet-smoothed categorical entropies and of the Gaussian differential entropies $\frac{1}{2}\log(2\pi e\, v_{c,j})$. For nominal attributes and $\alpha \to 0$ the bracket is $\sum_A \big[H(A) - H(A \mid C)\big] = \sum_A I(A; C)$, the information the partition carries about each attribute; the Gaussian terms are its analogue for fitted Gaussians. This is the information-theoretic category utility of Corter and Gluck [3], which the rational account reads as the mutual information between features and concepts [17].

Learning (`ifit`) sorts an instance from the root. At a leaf it either increments the leaf (exact match) or performs a *fringe split*: the leaf becomes an internal node with the old leaf and a new leaf for the instance as children. At an internal node it adds the instance and chooses, by category utility, between BEST (descend into the best child), NEW (create a leaf), MERGE (join the two best children under a new node) and SPLIT (replace the best child by its children).

A.2 Prediction

A *query* is an instance in which some attributes and some continuous coordinates are missing; we write O for the observed coordinates (missing continuous coordinates are encoded as `NaN`, missing attributes are absent). The log-likelihood of a query under a concept is $\ell_c(q) = \sum_{A \text{ obs.}} \log P_c(A{=}q_A) + \sum_{j \in O} \log p_c(q_j)$.

Definition A.3 (Multi-node prediction)**.** Prediction expands concepts best-first from the root, scoring each non-root concept c with one of

$$
\begin{aligned}
&\text{LIK}: && s_c = \ell_c(q),\\
&\text{PRIOR}: && s_c = \ell_c(q) + \log(n_c/n_{\text{root}}),\\
&\text{PATH}: && s_c = \pi_c(q) + \ell_c(q), \qquad \pi_c(q) = \textstyle\sum_{c' \in \text{path}(c)\setminus\{\text{root}\}} \log \frac{n_{c'} e^{\ell_{c'}(q)}}{\sum_{b \in \text{sib}(c')} n_b e^{\ell_b(q)}},\\
&\text{POSTERIOR}: && s_c = \pi_c(q),
\end{aligned}
$$

where $\text{sib}(c')$ are the children of c''s parent (including c'); under every weighting the root is scored by its likelihood, $s_{\text{root}} = \ell_{\text{root}}(q)$. With $E_M(q)$ the expanded set and $w_c(q) \propto e^{s_c}$ normalised over $E_M(q)$, the prediction of a missing continuous coordinate j and of a missing attribute A is

$$
\hat{y}_j(q) = \sum_{c \in E_M(q)} w_c(q)\, \mu_{c,j}, \qquad \hat{P}(A{=}v \mid q) = \sum_{c \in E_M(q)} w_c(q)\, P_c(A{=}v).
$$

The budget M bounds the number of expanded concepts. The *leaves-only* variant restricts the mixture to the expanded leaves; its budget counts leaves (expansion stops once M leaves have entered the mixture, after at most $64M$ expansions). *Full expansion* is $M = \infty$.

PATH is the default of the implementation and of `cobweb_torch`; $\pi_c(q)$ is the log posterior of c obtained by multiplying the sibling-normalised posteriors down its path, so $\sum_{c \text{ at depth } k} e^{\pi_c} \le 1$ for every depth k. The rational account [17] ranks concepts by pointwise mutual information $\log p(x \mid c) - \log p(x)$, i.e. by ℓ_c.

A.3 Cobweb forests

Definition A.4 (Messages)**.** Let $\mathcal{T}$ be a tree whose instances contain *outcome* attributes $V_{\mathcal{T}}$ (possibly none; Definition D.1). For a query q in which $V_{\mathcal{T}}$ is missing:

- the *activation* of $\mathcal{T}$ is its prediction of $V_{\mathcal{T}}$—the vector $\hat{y}_{V_{\mathcal{T}}}(q)$ for a continuous outcome, the predicted distribution $(\hat{P}(V_{\mathcal{T}}{=}v \mid q))_v$ for a nominal one;
- the *label* of $\mathcal{T}$ at depth k is the identifier of the depth-k concept on the root-to-leaf descent that chooses, at each node, the child maximising $\log n_c + \ell_c(q)$ (the leaf if the path is shorter; $k = \infty$ denotes the leaf).

Definition A.5 (Cobweb forest)**.** A Cobweb forest $\mathfrak{F} = (V, E, o)$ is a finite directed acyclic graph whose vertices are Cobweb trees and whose edges $\sigma \to \tau$ carry the activation and/or the label of σ into τ, together with an output tree $o \in V$. A tree τ learns instances made of (a) the raw coordinates x_{S_τ} it is wired to, (b) the activations of its parents as continuous coordinates, (c) the labels of its parents as nominal attributes (one attribute per parent), and (d) its outcome, if any. For an input x the forest computes the messages of every tree in topological order and outputs the activation of o. The *fan-in* of τ is the number of its non-outcome inputs (raw coordinates, activation coordinates, label attributes).

Definition A.6 (Training protocols)**.** Every tree learns only with `ifit`. Two protocols fix the messages a tree is trained on:

- *online*: for each observation i, every tree first emits its message for x_i from what it has learned from observations $1, \dots, i-1$, then learns observation i (activations are out-of-sample, as at test time; labels are concept identifiers, which can change as the tree restructures, so a child may store labels that its parent no longer emits);
- *split*: the stream is cut into consecutive chunks, one per level of the DAG; a tree at level k learns chunk k, reading the messages of its already-trained (frozen) parents.

Remark A.7 (Why not train stage-wise on the same data)*.* A tree that has learned (x_i, y_i) keeps a leaf for it (Lemma C.1), and when its variance floor is small compared with the spacing of the sample its activation at x_i is close to y_i itself. A parent trained on such in-sample messages learns the relation between y and a leaked copy of y, which it never sees at test time. The online protocol (predict, then learn), already used by the Cobweb language model, and the split protocol avoid this.

B Representation: statements and proofs

Throughout, a tree that *represents* $g : K \to \mathbb{R}$ ($K \subset \mathbb{R}^d$ compact) is a tree over instances $(x, g(x))$ queried with the last coordinate missing, and $\omega_g(r) = \sup\{|g(x) - g(x')| : \|x - x'\| \le r\}$ is the modulus of continuity.

B.1 A local error certificate

Lemma B.1 (Concept bias)**.** *Let c be a concept of a tree over instances $(x_i, g(x_i))$, with g L-Lipschitz on a convex set containing the instances' x-parts and the queries. Then $|\mu_{c,Y} - g(\mu_{c,X})| \le L\sqrt{\operatorname{tr} S_c}$, where S_c is the empirical covariance of the x-part of c's instances, and consequently, for every query x, every weighting and every budget for which the mixture is not empty,*

$$|\hat{y}(x) - g(x)| \;\le\; L \sum_{c \in E_M(x)} w_c(x) \Big(\|x - \mu_{c,X}\| + \sqrt{\operatorname{tr} S_c} \Big). \tag{3}$$

Proof. $|\mu_{c,Y} - g(\mu_{c,X})| = |\frac{1}{n_c}\sum_i (g(x_i) - g(\mu_{c,X}))| \le \frac{L}{n_c}\sum_i \|x_i - \mu_{c,X}\| \le L(\frac{1}{n_c}\sum_i \|x_i - \mu_{c,X}\|^2)^{1/2}$ by Jensen. For (3), write $\hat{y} - g(x) = \sum_c w_c(\mu_{c,Y} - g(\mu_{c,X})) + \sum_c w_c(g(\mu_{c,X}) - g(x))$ and use $\sum_c w_c = 1$. □

The right side of (3) is computable from the tree: a mixture is accurate when it puts its weight on concepts that are near the query *and* tight.

B.2 One tree

Theorem B.2 (A single tree is a universal approximator)**.** *Let $g \in C(K)$, $K \subset \mathbb{R}^d$ compact, and $\varepsilon > 0$. There is a Cobweb tree $\mathcal{T}$ and a variance floor $h^2 > 0$ such that $\sup_{x\in K} |\hat{y}_{\mathcal{T}}(x) - g(x)| < \varepsilon$ under full expansion with each of the weightings* LIK *and* PATH*, and with the leaves-only variant of any weighting. If g is L-Lipschitz and K lies in a cube of side D, a tree with G leaves achieves*

$$\sup_{x\in K} |\hat{y}_{\mathcal{T}}(x) - g(x)| \;\le\; C_d\, L\, D\, G^{-1/d} + C_d\, \|g\|_\infty\, G^{-1}. \tag{4}$$

Proof. Extend g continuously to a cube $Q \supseteq K$ of side D (Tietze's extension, or McShane's when g is L-Lipschitz, which keeps the constant L; its sup norm is at most $\|g\|_\infty + L\sqrt{d}\,D$, and the extra term is absorbed in the first term of (4)) and rescale Q to $[0,1]^d$, which multiplies the Lipschitz constant by D. Fix $\beta > 0$, let $\delta = \beta h$ and let Γ be the grid of cell centres of spacing δ, $G = |\Gamma| = \delta^{-d}$. Take the depth-one tree whose leaves are the instances $(\gamma, g(\gamma))$, $\gamma \in \Gamma$, and whose root pools them. Each leaf has $s^2 = 0$, hence variance h^2 in every coordinate; the root has variance at least $v_0 > 0$ (the variance of the grid) in every coordinate. Write $k_\gamma = (2\pi h^2)^{-d/2} e^{-\|x-\gamma\|^2/2h^2}$.

Leaves dominate. Some γ lies within $\sqrt{d}\,\delta/2$ of x, so $\max_\gamma k_\gamma \ge (2\pi h^2)^{-d/2} e^{-d\beta^2/8}$, while $\sum_\gamma k_\gamma \le (2\pi h^2)^{-d/2} C(\beta, d)$ by comparison with a Gaussian sum over a grid of spacing β in units of h. Under LIK the leaves carry unnormalised mass $\sum_\gamma k_\gamma \ge c\, h^{-d}$ and the root at most $(2\pi v_0)^{-d/2}$. Under PATH the leaf γ has $e^{\pi_\gamma} = k_\gamma / \sum_{\gamma'} k_{\gamma'}$ (equal counts), so its mass is $k_\gamma^2 / \sum k$, whose sum is at least $(\max k)^2 / \sum k \ge c' h^{-d}$, while the root again has mass at most $(2\pi v_0)^{-d/2}$. In both cases the root's weight is $O(h^d) = O(G^{-1})$.

Leaves form a kernel smoother. Among the leaves, $w_\gamma \propto k_\gamma^r$ with $r = 1$ (LIK) or $r = 2$ (PATH). For any $\rho > 0$, splitting the exponent and using the lower bound on $\max_\gamma k_\gamma^r$ for the normaliser,

$$\sum_{\|\gamma - x\| > \rho} w_\gamma \;\le\; \frac{e^{-r\rho^2/4h^2} \sum_\gamma e^{-r\|x-\gamma\|^2/4h^2}}{e^{-rd\beta^2/8}} \;\le\; C'' e^{-r\rho^2/4h^2},$$

so $|\sum_\gamma w_\gamma g(\gamma) - g(x)| \le \omega_g(\rho) + 2\|g\|_\infty C'' e^{-\rho^2/4h^2}$. Taking $\rho = \sqrt{h}$ makes this vanish as $h \to 0$, uniformly in x; adding the root's $O(h^d)$ weight (times $2\|g\|_\infty$) gives the first claim. For Lipschitz g bound the leaves' part directly: $|\sum_\gamma w_\gamma g(\gamma) - g(x)| \le LD \sum_\gamma w_\gamma \|\gamma - x\| \le C\, LD\, h$, because $\sum_\gamma e^{-r\|x-\gamma\|^2/2h^2} \|x - \gamma\| \le C(\beta, d)\, h$ by the same comparison with a Gaussian sum and the normaliser is bounded below as above; with $h \asymp G^{-1/d}$ and the root's weight $O(G^{-1})$ this is (4). The leaves-only variant drops the root term; restricted to the leaves of this tree, PRIOR coincides with LIK (equal counts) and POSTERIOR weights the leaves by $k_\gamma / \sum k$, i.e. $r = 1$. □

On the same flat tree the PRIOR weighting behaves differently: it multiplies each concept's likelihood by its *relative count*, and the root then keeps a share of the mixture that does not vanish as the variance floor shrinks.

Proposition B.3 (On a flat tree the prior- and posterior-weighted all-concept mixtures keep a fixed bias)**.** *In the setting of the proof of Theorem B.2, with* PRIOR *weighting and full expansion, the root's weight is $\lambda(x) = \mathcal{N}_{root}(x)/(\mathcal{N}_{root}(x) + \theta_h(x))$, where $\mathcal{N}_{root}(x) = \mathcal{N}(x; \mu_{root}, v_{root})$ and $\theta_h(x) = G^{-1} \sum_\gamma k_\gamma$ is bounded above by a constant $C(\beta, d)$ independent of h. Hence $\lambda(x) \ge \mathcal{N}_{root}(x)/(\mathcal{N}_{root}(x) + C(\beta, d))$ for all h, and $\hat{y}(x) - g(x) = \lambda(x)\big(\mu_{root,Y} - g(x)\big) + o(1)$: the error does not vanish wherever $g(x)$ differs from the grid mean of g. With* POSTERIOR *weighting the leaves' masses e^{π_γ} sum to exactly one, so $\lambda(x) = \mathcal{N}_{root}(x)/(\mathcal{N}_{root}(x) + 1)$ and the same conclusion holds.*

Proof. Leaf γ has prior $1/G$, so the leaves' mass is $\theta_h(x) = G^{-1}\sum_\gamma k_\gamma = \beta^d(2\pi)^{-d/2}\sum_\gamma e^{-\|x-\gamma\|^2/2h^2}$ (using $G^{-1} = \delta^d$), a Riemann sum of mesh β of the standard Gaussian density, bounded by $C(\beta, d)$ and bounded below by a positive constant (it is close to 1 in the interior when β is small, but it does not converge as $h \to 0$; it oscillates with x/h). The root has prior 1 and mass $\mathcal{N}_{\text{root}}(x)$, which tends to a positive limit. The leaves' mixture converges to $g(x)$ as in Theorem B.2. Under POSTERIOR the leaves are the root's children and their path posteriors are the sibling-normalised posteriors, which sum to one, while the root is scored by its likelihood (Definition A.3). □

Remark B.4 (A hierarchy changes the picture). The bias of Proposition B.3 belongs to the flat tree, not to the weightings alone. In a hierarchy each level of concepts carries a bounded share of the PRIOR mixture, so the root and the coarse levels together receive a share of order 1/depth: on a balanced binary tree over a grid of $G = 2^D$ points the PRIOR all-concept error falls roughly like $1/D$, i.e. like $1/\log G$, and in learned trees it falls like $1/\log N$ (Table 2). POSTERIOR behaves worse: soft sibling posteriors at the coarse splits leak mass across the split boundaries, and on the same binary tree its error stays at a floor even when the mixture is restricted to the leaves (Appendix H). The LIK and count-weighted leaves-only mixtures converge in both trees.

B.3 Composition: the Cobweb–Kolmogorov–Arnold theorem

Lemma B.5 (Approximating a DAG of continuous maps)**.** *Let D be a finite DAG of continuous maps $g_v : \mathbb{R}^{k_v} \to \mathbb{R}$ whose inputs are raw coordinates of $x \in [0,1]^n$ or outputs of other vertices, and let F be the map it computes; let K_v be the (compact) set of input vectors of v as x ranges over $[0,1]^n$. For every $\varepsilon > 0$ there are $\rho_v, \eta_v > 0$ such that any maps $\hat{g}_v$ with $\sup_{K_v + \bar{B}_{\rho_v}} |\hat{g}_v - g_v| \le \eta_v$ define a network $\hat{F}$ with $\sup_x |\hat{F}(x) - F(x)| \le \varepsilon$.*

Proof. Choose the tolerances in reverse topological order. At the output vertex put $\eta_o = \varepsilon/2$ and choose $\rho_o \le 1$ with $\omega_{g_o}(\rho_o) \le \varepsilon/2$ on the compact $K_o + \bar{B}_1$. Given the tolerance $\rho_v \le 1$ by which the input vector of v may deviate, allow each of its k_v inputs to deviate by $\rho_v/\sqrt{k_v}$: for each incoming vertex u set $\eta_u = \rho_v/(2\sqrt{k_v})$ and choose $\rho_u \le 1$ with $\omega_{g_u}(\rho_u) \le \rho_v/(2\sqrt{k_v})$ on $K_u + \bar{B}_1$, taking minima when u feeds several vertices. Then, in topological order, every vertex's input deviates from its exact value by at most ρ_v, so its output deviates by at most $\eta_v + \omega_{g_v}(\rho_v)$, which is within its consumers' tolerance; at the output the deviation is at most ε. □

Theorem B.6 (Cobweb–Kolmogorov–Arnold)**.** *Let $n \ge 2$. For every $f \in C([0,1]^n)$ and $\varepsilon > 0$ there is a Cobweb forest with activation messages, in which every tree has at most two inputs, consisting of*

- *$n(2n+1)$ univariate trees for the inner functions $\varphi_{q,p}$ (the functions do not depend on f; the resolution their trees need does),*
- *$(2n+1)(n-1)$ two-input trees forming the inner sums,*
- *$2n+1$ univariate trees for the outer functions Φ_q, and*
- *$2n$ two-input trees forming the outer sum,*

$4n^2 + 4n$ trees in all, whose output is within ε of f on $[0,1]^n$.

Proof. Write f in the form (2) and read it as a DAG of continuous maps of one or two variables: $\varphi_{q,p}$, the binary additions $(a,b) \mapsto a+b$ chained to form the $2n+1$ inner sums of n terms and the outer sum of $2n+1$ terms, and Φ_q. Apply Lemma B.5, and realise each $\hat{g}_v$ on the compact $K_v + \bar{B}_{\rho_v}$ by Theorem B.2 as a tree over instances $(\text{inputs}, g_v(\text{inputs}))$; its activation is the required $\hat{g}_v$. The count is $n(2n+1) + (2n+1)(n-1) + (2n+1) + 2n = 4n^2 + 4n$. □

Proposition B.7 (Two inputs are necessary)**.** *If every tree of a forest has fan-in at most one, its output depends on at most one raw coordinate. Hence fan-in two is the least fan-in at which Cobweb forests are universal.*

Proof. By induction in topological order, each tree's messages are functions of the at most one raw coordinate at the source of its unique input chain. □

Remark B.8 (Rates and smooth KA representations). Theorem B.6 uses a fixed number of trees, but for an arbitrary f the inner functions of (2) are rough, and the number of leaves each tree needs is not controlled. As for KANs [14, Thm. 2.1], the useful statement assumes structure. If f is computed by a DAG of s maps of fan-in $\le k$, each L-Lipschitz on a neighbourhood of its input set, then (4) and the stability argument of Lemma B.5 give a forest of s trees with G leaves each and error $C\,G^{-1/k}$, with C depending on s, k, L and the diameters of the input sets—a rate set by the fan-in k, not by n. A single tree over the n inputs has the upper bound $G^{-1/n}$ (Theorem B.2); we prove no matching lower bound, and a tree whose leaves pool examples can do better when f depends on fewer coordinates. Cobweb trees realise sums as learned two-input maps; KANs sum at nodes for free, which is why their rate is governed by one-dimensional splines.

B.4 Label forests

Theorem B.9 (Label forests are universal)**.** *For every $f \in C([0,1]^n)$ and $\varepsilon > 0$ there is a forest of n univariate trees without outcomes that pass leaf labels to an output tree whose instances consist of the n labels (nominal attributes) and the target, such that the output tree's prediction (LIK, leaves only, full expansion) is uniformly within ε of f.*

Proof. Let tree p be the depth-one tree over the K grid points of spacing $\delta = 1/K$ in $[0,1]$, all with count one and variance h^2. Its label for x_p is $\arg\max_\gamma\{\log n_\gamma + \ell_\gamma(x_p)\}$, the nearest grid point, so the labels partition $[0,1]^n$ into boxes B_t of side δ, indexed by label tuples t. Let the output tree be the depth-one tree with one leaf per tuple t, holding the instance $(t, f(\mathrm{centre}(B_t)))$. For a query with tuple t, a leaf t' that disagrees with t in m attributes has likelihood ratio $\big(\alpha/(1+\alpha)\big)^m$ relative to leaf t (Definition A.1, counts one). There are at most $\binom{n}{m}(K-1)^m$ such leaves, so their total weight is at most $\sum_{m\ge1}\binom{n}{m}(K\alpha)^m = (1+K\alpha)^n - 1$. Hence $|\hat y(x) - f(x)| \le \omega_f(\sqrt{n}\,\delta/2) + 2\|f\|_\infty\big((1+K\alpha)^n - 1\big)$, which is below ε for K large and then α small. □

The label forest of Theorem B.9 has an output tree with K^n leaves: labels carry no metric, so this construction pays for every cell of the product partition. Activations carry the metric and allow the fan-in-two construction of Theorem B.6.

Remark B.10 (Two elementary routes to universality). There are two short arguments that a single tree can represent any function, and the results above make each precise. The first is retrieval: store an example at every input on a fine grid, quantize the inputs, and predict the stored output of the query's cell—a nominal tree of grid cells does this, which is the construction of Theorem B.9. The second is mixtures: mixtures of enough Gaussians approximate any continuous density, and a tree's concepts are such a mixture. Approximating the joint density of (x, y) does not by itself make the *prediction* of y uniformly good, because the conditional mean divides by the marginal density of x, which can be small; Theorem B.2 therefore proves the regression statement directly, for the mixture weights the implementation actually uses, and Proposition B.3 shows that two of those weightings fail it on a flat tree.

B.5 Interweaving trees

Proposition B.11 (Merging trees under a common root gives mixtures)**.** *Let $\mathcal{T}_1, \dots, \mathcal{T}_m$ be trees over the same instance space with the same value sets (e.g. continuous-only trees), and let $\mathcal{T}$ be the tree whose root pools their roots' statistics and has $\mathcal{T}_1, \dots, \mathcal{T}_m$ as its subtrees. Under full expansion with* LIK *or* PATH *weighting,*

$$\hat{y}_{\mathcal{T}}(q) = \lambda_0(q)\,\mu_{root} + \sum_{j=1}^{m} \lambda_j(q)\,\hat{y}_{\mathcal{T}_j}(q), \qquad \lambda_j \geq 0,\ \sum_{j=0}^{m} \lambda_j = 1,$$

with $\lambda_0 \propto e^{\ell_{root}}$ and $\lambda_j \propto Z_j$ (LIK) or $\lambda_j \propto P(root_j \mid q)\, Z_j$ (PATH), where Z_j is the total unnormalised weight of $\mathcal{T}_j$'s own mixture and $P(root_j \mid q)$ the sibling-normalised posterior of $\mathcal{T}_j$'s root.

Proof. Under LIK a concept's score does not depend on the tree it sits in, so the mixture over $\mathcal{T}$ splits into the root term and the m subtree mixtures. Under PATH every concept of $\mathcal{T}_j$ gains the same term $\log P(\text{root}_j \mid q)$ in π_c (the sibling sets below root_j are unchanged), which rescales the subtree's weights uniformly and leaves its internal mixture unchanged. With different value sets the smoothing denominators K_A change, and the identity holds only approximately. □

Interweaving therefore closes the class of tree predictors under input-dependent convex combinations; it never composes them. Composition—one tree reading another's representation—is what forests add.

C Learning one tree: statements and proofs

Lemma C.1 (Leaf preservation)**.** *Run `ifit` on any sequence $z_1, \dots, z_N$ with any choices of operators (including add-only training), and suppose the depth limit is never reached. Then (i) every concept's statistics are the pooled statistics of the instances stored in the leaves below it; (ii) no leaf is ever deleted; (iii) every instance is stored in exactly one leaf, and a leaf stores an arriving instance only if the leaf's empirical variance is within the exact-match tolerance τ and the instance is within τ of the leaf's current mean (coordinatewise; nominal parts must have the same value proportions, within τ). With $\tau = 0$ a leaf stores copies of one instance, and if the instances are pairwise distinct the leaves are in bijection with them. (The implementation uses $\tau = 10^{-5}$ per coordinate.)*

Proof. By induction over the branches of the sorting loop. BEST increments the current node and descends; NEW increments the current node and creates a leaf holding z; MERGE inserts a new internal node whose statistics are the pooled statistics of the two children it adopts; SPLIT deletes the best child, which is internal (the operator requires it to have at least two children), and promotes its children; at a leaf, an exact match increments it and any other instance triggers a fringe split, which copies the leaf's statistics into a new internal parent, keeps the old leaf, and adds a leaf for z. No branch deletes a leaf, every instance ends in exactly one leaf, and a leaf only absorbs exact matches. Two copies of the same instance need not share a leaf: sorting may lead the second copy to a different leaf, which it then splits. □

When the depth limit is reached the implementation inserts the instance into the current node, which may be a leaf; (iii) then fails, which is why the lemma assumes the default limit (1000) is not reached.

Theorem C.2 (A learned tree is universally consistent)**.** *Let (X_i, Y_i) be i.i.d. with $X \in \mathbb{R}^d$, $|Y| \le B$, $m(x) = \mathbb{E}[Y \mid X = x]$. Let $\mathcal{T}_N$ be the tree that Cobweb learns from $(X_i, Y_i)_{i \le N}$ in any order and with any operator choices, with exact matching ($\tau = 0$) and without reaching the depth limit, and let it predict Y from X with the* count-weighted *leaves-only mixture (*PRIOR *weighting restricted to the leaves) under full expansion, with variance floor h_N^2. If $h_N \to 0$ and $N h_N^d \to \infty$, then $\int |\hat{y}_{\mathcal{T}_N}(x) - m(x)|\, \mu(dx) \to 0$ almost surely, whatever structure category utility builds (μ the distribution of X).*

Proof. By Lemma C.1 with $\tau = 0$, every instance is stored in exactly one leaf and a leaf c stores n_c copies of one instance (x_c, y_c), with zero spread. The PRIOR score of leaf c for a query x with Y missing is $\log \mathcal{N}(x; x_c, h_N^2 I) + \log(n_c/N)$, so the count-weighted leaves-only mixture is $\sum_c n_c K_{h_N}(x - x_c) y_c / \sum_c n_c K_{h_N}(x - x_c) = \sum_i K_{h_N}(x - X_i) Y_i / \sum_i K_{h_N}(x - X_i)$, the Nadaraya–Watson estimate with a Gaussian kernel. The Gaussian kernel is *regular* (it is bounded below by a positive constant on a ball and $\int \sup_{\|u-x\| \le r} K(u)\, dx < \infty$), and for regular kernels and bounded Y this estimate is strongly consistent in $L_1(\mu)$ under $h_N \to 0$, $N h_N^d \to \infty$ [8]; see Györfi et al. [11, Ch. 5] for the boxed-kernel version in L_2. □

Remark C.3 (Counts matter)*.* The leaves-only LIK mixture ignores counts. It equals the count-weighted mixture when every leaf stores one instance—for instance when X has a density and matching is exact—but not when instances repeat, exactly (atoms in a nominal input or in the target) or within the implementation's tolerance $\tau = 10^{-5}$: it then averages the distinct stored values instead of the samples. For a Bernoulli target whose copies share one leaf per value in each cell it gives $1/2$ in every cell as $\alpha \to 0$, whatever the cell's frequency (Appendix H). With $\tau > 0$ a leaf compares each arriving instance with its current mean, so the stored instances can drift from the leaf's mean by more than τ—of order $\tau \log n_c$ for a leaf with n_c copies; the count-weighted mixture is then the Nadaraya–Watson estimate on a design perturbed by that much. Since τ is fixed while $h_N \to 0$, Theorem C.2 describes the implementation only while $h_N \gg \tau \log N$, a pre-asymptotic regime that covers every experiment here except the smallest variance floors of Fig. 3. The PATH mixture of the experiments weights leaves by their path posteriors, which contain the counts.

Remark C.4 (What the structure is for)*.* Theorem C.2 holds because Cobweb never forgets an instance; the hierarchy plays no role in it. The structure matters for three things the theorem does not cover: (a) a *budget* $M < \infty$, where best-first search must find the leaves near the query; (b) the PATH weighting, where the mixture weight of a leaf is its path posterior times its likelihood, so the leaves-only mixture is not a kernel smoother in a learned tree (Lemma B.1 still bounds its error); and (c) the *labels* a tree passes to other trees, which are its concepts. The experiments use the implementation's default, PATH with a leaf budget; they are therefore a test of these structural effects as much as of the theorem. The branch-and-bound stop of `categorize(use_best=True)` ends the whole search as soon as the best score exceeds the path posterior of the concept just expanded. This is sound only if no concept still in the queue has a larger path posterior, but the queue is ordered by score, not by path posterior; and with Gaussian densities scores can exceed path posteriors because $\ell_c > 0$. The stop can therefore return a concept whose score is below the best, with continuous attributes and with nominal ones alike (Appendix H exhibits both). A sound stop must bound what a descendant can still score: with $\ell_{\max}$ an upper bound on any concept's log-likelihood (0 for nominal attributes, $-\frac{|O|}{2}\log(2\pi h^2)$ for $|O|$ observed Gaussian coordinates), a subtree whose root has $\pi_c + \ell_{\max}$ below the best score found so far can be pruned.

D Outcomes and tasks: statements and proofs

A Cobweb tree is unsupervised: it models the joint distribution of whatever its instances contain and predicts any attribute that is missing from a query. What a tree *passes on* is therefore fixed by what its instances contain. We make this the central object.

D.1 Tasks as inputs and outcomes

Definition D.1 (Task)**.** Let the stream consist of i.i.d. observations (X, Y), $X \in \mathbb{R}^n$ the raw input and Y the target. A *task* for a tree τ is a pair (C_τ, V_τ):

- the *input* C_τ, what the tree perceives: its raw coordinates X_{S_τ} and the messages of its parents;
- the *outcome* V_τ, what the tree predicts and later observes: either nothing ($V_\tau = \varnothing$) or an *observable* variable, i.e. coordinates of the observation that the input does not contain—the target Y, or raw coordinates X_J with $J \cap S_\tau = \emptyset$.

The tree learns the instances (C_τ, V_τ) with `ifit` and is queried with V_τ missing (Definition A.4). A *task assignment* A gives every tree of a forest a task; the output tree's outcome is Y.

Three families of tasks occur in the systems that motivate this paper. *Categorization* ($V_\tau = \varnothing$): the tree passes only labels, as the filter hierarchy of Convolutional Cobweb does. *Funnel* ($V_\tau = Y$): every tree predicts the final target, as every tree of the Cobweb language model predicts the next word. *Cross-view* ($V_\tau = X_j$, $j \notin S_\tau$): the tree predicts a part of the input it does not perceive—a neighbouring pixel, the next token, the context of a chunk, as TRELLIS's content hierarchy models its context. The online protocol of Definition A.6 is "predict, then observe, then learn".

Observability is the constraint that separates learnable from representable forests.

Remark D.2 (The KA tasks are not observable). In Theorem B.6 the tree for Φ_q holds the values $\Phi_q(u_q)$. The stream never shows them: it shows only their sum $Y = \sum_q \Phi_q(u_q)$. A KAN separates the terms of the sum by back-propagating the error of the sum through each of them; a Cobweb tree can only predict what it will observe. The inner functions $\varphi_{q,p}$ do not depend on f, so a forest could be *given* them as fixed tasks, but the outer decomposition is not identifiable from the stream without further structure. The rest of this section asks which *observable* tasks recover f anyway.

D.2 What a learned forest converges to

Definition D.3 (Ideal forest, sufficiency, deficit)**.** Let A be a task assignment of an activation forest. Define, in topological order, the *ideal input* and *ideal message* of each tree,

$$U_\tau = \big(X_{S_\tau}, (m_\sigma(X))_{\sigma \to \tau}\big), \qquad m_\tau(X) = \mathbb{E}[V_\tau \mid U_\tau] = \psi_\tau(U_\tau),$$

and the *ideal output* $F_A(X) = m_o(X) = \mathbb{E}[Y \mid U_o]$. Let $f(X) = \mathbb{E}[Y \mid X]$. The assignment is *sufficient* if $F_A = f$ almost surely, and its *deficit* is

$$\Delta(A) = \mathbb{E}\big[(f(X) - F_A(X))^2\big] = \mathbb{E}\,\mathrm{Var}(Y \mid U_o) - \mathbb{E}\,\mathrm{Var}(Y \mid X).$$

The identity holds because U_o is a function of X, so $F_A = \mathbb{E}[f(X) \mid U_o]$ and $\mathrm{Var}(Y \mid U_o) = \mathbb{E}[\mathrm{Var}(Y \mid X) \mid U_o] + \mathrm{Var}(f(X) \mid U_o)$.

Assumption D.4 (Regularity)**.** (A1) The observations are i.i.d., X has compact support, and every outcome is bounded, $|V_\tau| \le B$. (A2) For every tree, the ideal input $U_\tau \in \mathbb{R}^{k_\tau}$ has a density bounded away from zero and infinity on its support D_τ, a compact set with Lipschitz boundary, and ψ_τ extends to a uniformly continuous function on a neighbourhood of D_τ. (A3) Every tree matches instances exactly ($\tau = 0$) and predicts with the count-weighted leaves-only mixture under full expansion, with variance floor $h_{\tau,N}^2$, $h_{\tau,N} \to 0$ and $N h_{\tau,N}^{k_\tau} / \log N \to \infty$. (A4) For every edge $\sigma \to \tau$, $\varepsilon_{\sigma,N}/h_{\tau,N} \to 0$ in probability, where $\varepsilon_{\sigma,N} = \sup_x |\hat{m}_\sigma(x) - m_\sigma(x)|$ is the parent's uniform error. (A5) For every tree, the Nadaraya–Watson estimate with the Gaussian kernel and bandwidth $h_{\tau,N}$, built on N i.i.d. ideal inputs U_τ and outcomes, converges to ψ_τ uniformly on the δ_N-neighbourhood of D_τ in probability, for every $\delta_N = o(h_{\tau,N})$, and the normalised kernel mass $N^{-1}\sum_i K_h(u - U_i)$ is bounded below by a positive constant uniformly on that neighbourhood, with probability tending to one.

(A5) is a uniform law for kernel smoothers up to the boundary. The cited results [7, 15] state it on interior compact sets; up to a Lipschitz boundary it is expected to follow from the cone condition (a fixed fraction of every small ball around a point of D_τ lies in D_τ), the uniform kernel-density law of Einmahl and Mason [9] and the bias bound $\omega_{\psi_\tau}(Rh) + 2Be^{-R^2/16}$; we state it as an assumption rather than derive it.

(A4) is a condition on the schedule: by induction the parents' errors tend to zero (Theorem D.5), so variance floors that shrink slowly enough satisfy it. (A2) is restrictive: an ideal message with a critical point, such as $\mathbb{E}[Y \mid X_2] \propto e^{X_2^2}$ at $X_2 = 0$, has an unbounded density there, so the funnel forests of the KAN examples are outside Theorem D.5; the experiments test them anyway.

Theorem D.5 (A Cobweb-learned forest converges to its ideal forest)**.** *Let $\mathfrak{F}$ be an activation forest with task assignment A, trained with the split protocol on N observations, and let Assumption D.4 hold. Then for every tree τ, $\sup_x |\hat{m}_\tau(x) - m_\tau(x)| \to 0$ in probability; in particular the output satisfies $\sup_x |\hat{F}_N(x) - F_A(x)| \to 0$ in probability.*

Proof. Induction over the levels of the DAG. Fix a tree τ at level ℓ and condition on the chunks of the earlier levels, on which its parents were trained and frozen. Write $\hat{u}_\tau(x) = (x_{S_\tau}, (\hat{m}_\sigma(x))_\sigma)$ for its learned input map and $u_\tau(x)$ for the ideal one, so $\|\hat{u}_\tau - u_\tau\|_\infty \le \varepsilon := \sqrt{k_\tau}\max_\sigma \varepsilon_{\sigma,N}$.

Step 1: the tree is a kernel smoother. The tree learns the instances $(\hat{U}_i, V_i) = (\hat{u}_\tau(X_i), V_{\tau,i})$ of chunk ℓ, which are i.i.d. given the earlier chunks. By Lemma C.1 its leaves store copies of single instances, with zero spread, and, as in the proof of Theorem C.2, its count-weighted leaves-only activation at a input u is the Nadaraya–Watson estimate $\hat{\psi}(u) = \sum_i K_h(u - \hat{U}_i)V_i / \sum_i K_h(u - \hat{U}_i)$ over the instances of the chunk, with the Gaussian kernel of bandwidth $h = h_{\tau,N}$. (With the implementation's tolerance the stored inputs move by at most about $\tau(1 + \ln N)$ (Remark C.3), a perturbation that Step 2 absorbs when $\tau \ln N = o(h)$.)

Step 2: the perturbed design does not matter. Let $\check{\psi}$ be the same estimate built on the ideal inputs $U_i = u_\tau(X_i)$. By (A5), $\check{\psi} \to \psi_\tau$ uniformly on the δ-neighbourhood of D_τ for any $\delta = o(h)$, in probability. Write $\hat{U}_i = U_i + \Delta_i$, $\|\Delta_i\| \le \varepsilon$. For $\|u - U_i\| \le Rh$,

$$\frac{K_h(u - \hat{U}_i)}{K_h(u - U_i)} = \exp\Big(\frac{2\langle u - U_i, \Delta_i\rangle - \|\Delta_i\|^2}{2h^2}\Big) \in \big[e^{-\eta}, e^{\eta}\big], \qquad \eta = \frac{R\varepsilon}{h} + \frac{\varepsilon^2}{2h^2},$$

and for $\|u - U_i\| > Rh \ge 2\varepsilon$, $K_h(u - \hat{U}_i) \le (2\pi h^2)^{-k/2} e^{-\|u - U_i\|^2/8h^2}$. Since the density of U_τ is bounded below on D_τ and bounded above, and D_τ has Lipschitz boundary, the kernel mass of the near points is at least $c\,N$ and that of the far points at most $C\,N e^{-R^2/16}$, uniformly in u within $o(h)$ of D_τ, with probability tending to one, by the uniform consistency of kernel density estimates

[9]. Hence $\sup_u |\hat{\psi}(u) - \check{\psi}(u)| \le 2B\big(e^{2\eta} - 1 + C' e^{-R^2/16}\big)$, which tends to zero in probability when $R = R_N \to \infty$ slowly enough that $R\varepsilon/h \to 0$, possible by (A4).

Step 3: the test-time input. At test time the tree is queried at $\hat{u}_\tau(x)$, so $|\hat{m}_\tau(x) - m_\tau(x)| \le \sup_{D_\tau^{(\varepsilon)}} |\hat{\psi} - \psi_\tau| + \omega_{\psi_\tau}(\varepsilon)$, where $D_\tau^{(\varepsilon)}$ is the ε-neighbourhood of D_τ; both terms tend to zero in probability by Steps 1–2 and (A2). This is the induction hypothesis for the children of τ; at the output tree it is the claim. □

Corollary D.6 (Consistency if and only if sufficiency)**.** *Under the hypotheses of Theorem D.5, $\mathbb{E}[(\hat{F}_N(X) - f(X))^2] \to \Delta(A)$. The forest is consistent for f if and only if its task assignment is sufficient.*

Proof. $\hat{F}_N$ and f are bounded by B, so uniform convergence in probability gives $\mathbb{E}(\hat{F}_N - F_A)^2 \to 0$, and $\mathbb{E}(\hat{F}_N - f)^2 \to \mathbb{E}(F_A - f)^2 = \Delta(A)$. □

Theorem D.5 separates what Cobweb's learning does from what the tasks determine: every tree learns the conditional expectation of its outcome given its input, nothing more and nothing less; whether the composition of these conditional expectations is f is a property of the task assignment alone.

D.3 Which tasks are sufficient

Funnel tasks. A convenient sufficient condition for funnel forests is that each tree's ideal message equals $\mathbb{E}[Y \mid X_{R(\tau)}]$, where $R(\tau)$ is the set of raw coordinates upstream of τ; this holds by induction when every $\mathbb{E}[Y \mid X_{R(\tau)}]$ is a function of the parents' $\mathbb{E}[Y \mid X_{R(\sigma)}]$ and of X_{S_τ}.

Proposition D.7 (Funnel tasks are sufficient for monotone-additive targets)**.** *Let X have compact support and consist of independent blocks $X_{B_1}, \dots, X_{B_m}$, and let $Y = \Phi\big(\sum_j g_j(X_{B_j})\big) + \xi$ with Φ continuous and strictly increasing, g_j continuous, and ξ centred and independent of X. Then every funnel activation forest whose graph is a rooted tree in which each block is the raw input of exactly one tree is sufficient.*

Proof. For an upstream set of blocks S, $\mathbb{E}[Y \mid X_S] = \mathbb{E}[\Phi(s_S + Z_{\bar{S}})] =: H_S(s_S)$ with $s_S = \sum_{j \in S} g_j(X_{B_j})$ and $Z_{\bar{S}}$ the sum over the other blocks, independent of X_S; H_S is continuous and strictly increasing because Φ is. A tree whose children cover the disjoint block sets $S_1, \dots, S_k$ and which reads the raw blocks S_0 has $s_S = \sum_i H_{S_i}^{-1}(m_{S_i}) + \sum_{j \in S_0} g_j(X_{B_j})$, so its $\mathbb{E}[Y \mid X_S] = H_S(s_S)$ is a continuous function of its parents' messages and its raw input. □

The KAN examples $\exp(\sin \pi x_1 + x_2^2)$, $\exp\big(\frac{1}{100}\sum_i \sin^2(\pi x_i/2)\big)$ and $\exp\big(\frac{1}{2}(\sin \pi(x_1^2 + x_2^2) + \sin \pi(x_3^2 + x_4^2))\big)$ (the last with the blocks (x_1, x_2) and (x_3, x_4)) are in this class.

Proposition D.8 (Funnel tasks can fail)**.** *Let X_1, X_2 be independent and symmetric about 0, with densities bounded away from zero and infinity on compact intervals, and $Y = X_1 X_2$. For the funnel forest whose output tree reads the activations of a tree perceiving X_1 and a tree perceiving X_2, the ideal messages are $m_1 = m_2 \equiv 0$, so $F_A \equiv 0$ and $\Delta(A) = \mathrm{Var}(Y)$. If the trees are learned as in Theorem D.5, with variance floors satisfying (A3) and, for the output tree, (A4), the learned output converges uniformly in probability to $\mathbb{E}Y = 0$: in the limit the forest learns nothing, whatever structure its trees build.*

Proof. $\mathbb{E}[X_1 X_2 \mid X_1] = X_1 \mathbb{E} X_2 = 0$, and likewise for X_2, so the output tree's ideal input is the constant $u_0 = (0, 0)$. The source trees satisfy Assumption D.4, so their activations converge to 0 uniformly (Steps 1–3 of the proof of Theorem D.5). The output tree therefore learns and is queried

at inputs within $\varepsilon \to 0$ of u_0; all its kernel weights lie within the factor $e^{-2\varepsilon^2/h^2} \to 1$ of one another, so its prediction converges to the mean of Y over its chunk, and that mean to $\mathbb{E}Y = 0$. At finite N the activations are noisy functions of the coordinates and can carry a little information; the statement is asymptotic. □

Cross-view tasks. A tree that predicts another part of the input never sees the target. It recovers what the views share, which suffices when the target depends on the input only through what the views share—the multi-view redundancy assumption of co-training and of contrastive learning [2, 23].

Proposition D.9 (Cross-view tasks are sufficient under multi-view redundancy)**.** *Let $Z \in \mathbb{R}^r$ be latent and $X = (X^{(1)}, \dots, X^{(m)})$ consist of $m \geq 2$ views with (Z, X) jointly Gaussian and centred, $X^{(v)} = \Lambda_v Z + E_v$, $\mathrm{Cov}(Z)$ and every $\mathrm{Cov}(E_v)$ nonsingular, the noises E_v independent of each other and of Z. Let $Y = g(Z) + \xi$ with $\mathbb{E}|g(Z)| < \infty$ and ξ centred and independent of (Z, X). Give the tree of view v the input $X^{(v)}$ and as outcome r coordinates W_v of other views whose loading matrix $\Lambda_{W_v} \in \mathbb{R}^{r \times r}$ is invertible, and let the output tree have the activations of the m view trees as input and Y as outcome. This assignment is sufficient, and no tree but the output tree perceives Y.*

Proof. Write $\mu_v = \mathbb{E}[Z \mid X^{(v)}]$ and P_v for the posterior precision given $X^{(v)}$, and P_0 for the prior precision; both precisions are constants. Since $W_v = \Lambda_{W_v} Z + E'$ with E' independent of $(Z, X^{(v)})$, the ideal message is $m_v = \mathbb{E}[W_v \mid X^{(v)}] = \Lambda_{W_v} \mu_v$, an invertible linear function of μ_v. By conditional independence of the views, $p(z \mid x) \propto p(z) \prod_v p(x^{(v)} \mid z) \propto p(z)^{1-m} \prod_v p(z \mid x^{(v)})$, a Gaussian with precision $P = \sum_v P_v - (m-1)P_0$ and mean $P^{-1} \sum_v P_v \mu_v$. The posterior of Z given X therefore depends on X only through $(\mu_v)_v$, hence through $(m_v)_v$, and so does $f(X) = \mathbb{E}[g(Z) \mid X]$. Thus $F_A = \mathbb{E}[Y \mid (m_v)_v] = f$. □

The assumption is necessary in the following sense: if the target carries a view-specific component, e.g. $Y = e^\top E_1$ with $X^{(1)}$ of dimension larger than r and e generic, the cross-view messages discard it and the deficit is positive. In images and text the views are patches and contexts, and the cross-view outcome is a neighbouring pixel or the next token.

Categorization. A tree without an outcome passes labels. Its labels do not depend on f at all, which is their strength and their cost.

Theorem D.10 (Consistency of label forests)**.** *Let a forest consist of n trees without outcomes over the raw coordinates passing labels at depth k_N (or leaf labels) to an output tree that models (labels, Y), $|Y| \leq B$, learned with the split protocol; let the output tree match instances exactly and predict with the count-weighted leaves-only mixture under full expansion. Let X have compact support $\mathcal{X}$, let $\mathcal{P}_N$ be the product partition induced by the frozen label maps, and let M_N be the number of its cells that meet $\mathcal{X}$. If, in probability, (a) $\max_C \mathrm{diam}(C \cap \mathcal{X}) \to 0$ over the cells C meeting $\mathcal{X}$ (the label maps partition all of the input space, so cells at the edge are unbounded; only their part in $\mathcal{X}$ matters), (b) $M_N/N \to 0$, and (c) $\alpha_N M_N \to 0$, then the forest's prediction $\hat{y}$ satisfies $\mathbb{E} \int (\hat{y} - m)^2 \, d\mu \to 0$.*

Proof. Condition on the frozen partition and let $N' \asymp N$ be the size of the output chunk, n_C the number of its points in cell C, and $\hat{m}$ the partitioning estimate ($\hat{m} = \bar{y}_C$, the mean of the chunk's Y in C, and $\hat{m} = 0$ on empty cells). Under (a) and (b) the proof of Györfi et al. [11, Thm. 4.2], applied to the fixed partition (its conditions are stated for the cells meeting a ball, which here may be taken to contain $\mathcal{X}$), gives $\mathbb{E} \int (\hat{m} - m)^2 d\mu \to 0$; it remains to show $\mathbb{E} \int (\hat{y} - \hat{m})^2 d\mu \to 0$.

Take a query in a cell C with $n_C \geq 1$ and label tuple t_C. By Lemma C.1 ($\tau = 0$) every leaf c stores n_c copies of one instance (t_c, y_c), and the leaves with $t_c = t_C$ store exactly the n_C points of C. With K_a labels for tree a, an attribute that a leaf matches has probability $(n_c + \alpha)/(n_c + (K_a + 1)\alpha) \in [\kappa_a, 1]$ with $\kappa_a = (1 + (K_a + 1)\alpha)^{-1}$, and one it does not match has probability $\alpha/(n_c + (K_a + 1)\alpha) \leq \alpha/n_c$. Hence, with $\kappa = \prod_a \kappa_a$, the matching leaves carry count-weighted mass between κn_C and n_C, distributed in proportion to their counts up to factors in $[\kappa, 1]$, and every other leaf carries at most $n_c(\alpha/n_c)^m \leq \alpha$, where $m \geq 1$ is the number of attributes it misses. There are at most N' other leaves, so

$$|\hat{y} - \bar{y}_C| \;\leq\; 2B\Big(\min\Big(1, \frac{\alpha N'}{\kappa n_C}\Big) + \frac{1-\kappa}{\kappa}\Big),$$

and $|\hat{y} - \hat{m}| \leq B$ when $n_C = 0$. Every label of tree a is the label of some cell, so $K_a \leq M_N$ and $1 - \kappa \leq \alpha \sum_a (K_a + 1) \leq n\alpha(M_N + 1) \to 0$ by (c). For a cell of probability $\mu(C)$, $\Pr(n_C < N'\mu(C)/2) \leq e^{-N'\mu(C)/8}$ (Chernoff), so

$$\mathbb{E}\int \min\Big(1, \frac{\alpha N'}{\kappa n_C}\Big) d\mu \;\leq\; \sum_{C \cap \mathcal{X} \neq \emptyset} \mu(C)\Big(e^{-N'\mu(C)/8} + \frac{2\alpha}{\kappa\,\mu(C)}\Big) \;\leq\; \frac{8M_N}{eN'} + \frac{2\alpha M_N}{\kappa},$$

using $te^{-N't/8} \leq 8/(eN')$; both terms tend to zero by (b) and (c). Therefore $\mathbb{E}\int(\hat{y} - \hat{m})^2 d\mu \to 0$. The bounds hold conditionally on the partition; since everything is bounded, the conditions in probability and bounded convergence give the claim. □

Remark D.11 (The trade-off). Condition (b) requires the product partition to have fewer occupied cells than samples; if X fills the K^n cells of K labels per tree, that is $K^n = o(N)$, so the rate of a label forest is governed by n, not by the fan-in. For funnel and cross-view forests Theorem D.5 gives consistency but no rate; the bounds of Remark B.8, which are for constructed trees, suggest rates governed by the fan-in, at the price of sufficiency (Proposition D.8).

Conjecture D.12 (Hybrid messages)**.** *A forest whose trees pass labels and funnel activations, and whose output tree's variance floor on the activation coordinates satisfies an (A3)–(A4)-type schedule, is consistent whenever its label cells satisfy the conditions of Theorem D.10 (through its labels), and attains the fan-in rate when its funnel assignment is sufficient (through its activations). (Without a schedule on the floor the conjecture fails: as $h_N \to 0$ quickly the output tree becomes a nearest-neighbour rule in activation space and ignores the labels.)*

Remark D.13 (When should a forest beat a single tree?). Comparing the upper bounds suggests where to look, without proving a separation. For a noise-free C^2 target on an i.i.d. design, a single Nadaraya–Watson tree's interior error is of order N^{-1} for $n = 1$, $N^{-1/2}$ for $n = 2$ and $N^{-2/(n+2)}$ for $n \geq 3$ (the design itself fluctuates; on a compact support the $O(h)$ bias in a boundary layer of width $O(h)$ lowers every exponent but leaves the comparison below unchanged); each tree of a funnel forest regresses an outcome that is noisy given its own input (the coordinates it does not perceive act as noise, of variance $\mathrm{Var}(Y \mid X_{R(\tau)})$), with error of order $N^{-2/(k+4)}$. The bounds favour the forest only when $n > k + 2$. For a noisy target the single tree's bound is $N^{-2/(n+4)}$ and the comparison favours the forest as soon as $n > k$.

E Choosing tasks: statements and proofs

Corollary D.6 reduces learnability to choosing a sufficient task assignment. The assignment is part of the forest's structure; we ask whether it can be inferred from the stream, and by what rule.

E.1 Tasks cannot be inferred locally

Definition E.1 (Local task rule)**.** A *local task rule* decides something about a tree τ—keep or prune it, which task to give it, which message to pass—as a measurable function of τ's own training stream (the instances $(C_{\tau,i}, V_{\tau,i})$, $i \le N$, for a candidate outcome V_τ, possibly $\varnothing$) and of independent randomness (in the implementation each tree of a forest has its own tie-breaking stream). The tree's own prequential score, its category utility and its learned concepts are all functions of this stream.

Theorem E.2 (Locality obstruction)**.** *Let X_1, X_2, X_3 be independent and uniform on $[-1, 1]$, observe only (X_1, X_2, Y), and consider the two worlds*

$$P_a : \; Y = \operatorname{sign}(X_1 X_2), \qquad P_b : \; Y = \operatorname{sign}(X_1 X_3).$$

(i) For $j \in \{1, 2\}$, the pair (X_j, Y) has the same law under P_a and P_b: X_j and Y are independent and Y is uniform on $\{-1, 1\}$. Hence, for every candidate outcome, the training stream of a tree with input X_j has the same law in both worlds, and so does the output of every local task rule. (ii) Under P_a, $f(X) = Y$ and discarding X_2 loses everything ($\mathbb{E}\operatorname{Var}(Y \mid X_1) = 1$ while $\mathbb{E}\operatorname{Var}(Y \mid X) = 0$); under P_b, X_2 is irrelevant ($f(X) = \mathbb{E}[Y \mid X_1] = 0$). Consequently every local rule that decides whether to keep the tree perceiving X_2 errs with probability at least $1/2$ in one of the two worlds. (iii) Under P_a the funnel activations of both univariate trees vanish, so every funnel forest built on them has the maximal deficit $\Delta = \operatorname{Var}(Y) = 1$, whereas the label forest of Theorem D.10 and a single tree over (X_1, X_2) are sufficient.

Proof. (i) Under P_a, $Y = \operatorname{sign}(X_1)\operatorname{sign}(X_2)$; given X_2, the factor $\operatorname{sign}(X_1)$ is uniform on $\{-1, 1\}$ and independent of X_2, so Y is uniform and independent of X_2; by symmetry the same holds for X_1. Under P_b, $Y = \operatorname{sign}(X_1)\operatorname{sign}(X_3)$ is independent of X_2 outright and, by the same argument, of X_1. The stream of a tree with input X_j is an i.i.d. sample of (X_j, V) with $V \in \{Y, \varnothing\}$ (or another observable whose joint law with X_j is the same in both worlds, such as X_{3-j}), so it has the same law in both worlds, and so does any measurable function of it. (Definition D.1 lets an outcome be Y or raw coordinates, not both; a tree whose stream contained X_j, X_{3-j} and Y together would see everything observed, and telling the worlds apart from that stream is composition.) (ii) Direct. If p is the probability that the rule keeps the tree, it errs with probability $1 - p$ under P_a and p under P_b. (iii) $\mathbb{E}[Y \mid X_j] = 0$ for $j = 1, 2$ by (i), so Proposition D.8's argument applies; the label and single-tree statements follow from Theorems C.2 and D.10. □

Corollary E.3 (Parity requires composition)**.** *Let $X_1, \ldots, X_n$ be independent and uniform on $[-1, 1]$ and $Y = \prod_{j \in J} \operatorname{sign}(X_j)$. For every S that does not contain J, Y is independent of X_S. Hence no tree whose upstream raw set $R(\tau)$ misses a coordinate of J, and no rule using the streams of any set of trees whose inputs and outcomes together are functions of X_S with $J \not\subseteq S$, can detect the dependence; every funnel activation of a tree with $J \not\subseteq R(\tau)$ vanishes; and the ideal output of an activation forest whose outcomes are Y or raw coordinates equals f only if the raw input of some tree contains J (a merged input). Labels, which pass a tree's input rather than a prediction, are the other way to bring all of J to one tree; for learned forests these statements hold in the limit of Theorem D.5.*

Proof. For $j \in J \setminus S$, $\operatorname{sign}(X_j)$ is uniform on $\{-1, 1\}$ and independent of X_S and of the other factors, so Y is uniform and independent of X_S. For the last claim, suppose no tree's raw input contains J and argue by induction in topological order that every ideal activation is constant. If a tree's parents' ideal activations are constant, its ideal input is a function of X_{S_τ} and its ideal message is $\mathbb{E}[V_\tau \mid X_{S_\tau}]$. For $V_\tau = Y$ this is $\mathbb{E}Y$, because $J \not\subseteq S_\tau$; for $V_\tau = X_j$ with $j \notin S_\tau$ it is $\mathbb{E}X_j$,

because the coordinates are independent. Hence the output's ideal value is $\mathbb{E}[Y \mid X_{S_o}] = \mathbb{E}Y = 0$, while $f(X) = Y$ is not constant. □

Theorem E.2 is not a peculiarity of Cobweb: synergistic information—information about Y held by a set of variables jointly and by none of them alone [25]—is invisible to any module that sees one of them. Back-propagation escapes it because each module receives the error of the composition. A Cobweb tree receives no error, so the decision has to be taken where the information is composed.

E.2 Verification at the point of composition

Theorem E.4 (Task selection by verification)**.** *Let $|Y| \le B$ and let $\mathcal{A}$ be a finite set of task assignments. For each $A \in \mathcal{A}$ let $\hat{F}_{A,N}$ be the output of the forest that Cobweb learns with assignment A from a training stream of length N, with $|\hat{F}_{A,N}| \le B$ and $\mathbb{E}\int |\hat{F}_{A,N} - F_A|^2 d\mu \to 0$ (Theorems C.2, D.5 and D.10 give this for activation forests, label forests and single trees; for a single tree, and for a label forest under the conditions of Theorem D.10, $F_A = f$). Let $\hat{A}_N$ minimise the mean squared error of Y on an independent validation stream of length $N_v(N) \to \infty$. Then*

$$\Pr\Big[\Delta(\hat{A}_N) = \min_{A \in \mathcal{A}} \Delta(A)\Big] \to 1.$$

In particular, if $\mathcal{A}$ contains a sufficient assignment, the selected assignment is sufficient with probability tending to one and the selected forest is consistent.

Proof. Let $\sigma^2 = \mathbb{E}\,\mathrm{Var}(Y \mid X)$. Conditionally on the training stream, the expected validation error of A is $R_N(A) = \sigma^2 + \mathbb{E}[(f - \hat{F}_{A,N})^2 \mid \text{train}]$, and $\mathbb{E}[(f - \hat{F}_{A,N})^2 \mid \text{train}] \to \Delta(A)$ in probability, because $\|f - \hat{F}_{A,N}\|_{L_2(\mu)} - \|f - F_A\|_{L_2(\mu)}$ is bounded by $\|\hat{F}_{A,N} - F_A\|_{L_2(\mu)} \to 0$. The validation average of the bounded squared errors differs from $R_N(A)$ by $O_P(N_v^{-1/2})$ (Chebyshev). With $\gamma = \min\{\Delta(A) - \Delta^* : \Delta(A) > \Delta^*\} > 0$, $\Delta^* = \min_A \Delta(A)$, every assignment with $\Delta(A) > \Delta^*$ has a larger validation error than every minimiser with probability tending to one, and $\mathcal{A}$ is finite. □

The validation error at the output is the outcome of the composing tree. With the log loss in place of the squared error the same rule would estimate the conditional entropy $H(Y \mid U_o)$ and select the task assignment that maximises the predictive information $I(Y; U_o)$; proving this needs the convergence of the trees' predictive distributions, not only of their means, which we leave open. Category utility measures the same kind of quantity—the information a partition carries about an attribute—when Cobweb chooses among its operators; task selection evaluates it one level up, for the outcome and the composing tree's input. In the online protocol the prequential error $\frac{1}{N}\sum_i (Y_i - \hat{F}_{A,i-1}(X_i))^2$ plays the role of the validation error [5]; whenever the online forest's risk tends to that of F_A, its expectation converges to the same limit by a Cesàro argument, and the average itself converges almost surely because $(Y_i - \hat{F}_{A,i-1}(X_i))^2$ minus its conditional expectation given the past is a bounded martingale difference sequence, to which the strong law for martingales applies.

E.3 Task operators

Definition E.5 (Task operators and synergy deficit)**.** Let a tree with outcome V compose two subtrees σ_1, σ_2 with inputs C_1, C_2. The *task operators* at this composition are

- ACT: σ_j gets the outcome V and passes its activation;
- LABEL: σ_j gets no outcome and passes its label;

- HYBRID: σ_j gets the outcome V and passes its activation and label;
- CROSS: σ_j gets a coordinate of C_{3-j} as outcome and passes its activation;
- MERGE: σ_1, σ_2 are replaced by one tree with input $C_1 \cup C_2$.

The *synergy deficit* of the pair is $\Delta_{\text{ACT}}(\sigma_1, \sigma_2) = \mathbb{E}\,\mathrm{Var}(V \mid a_1, a_2) - \mathbb{E}\,\mathrm{Var}(V \mid C_1, C_2) \geq 0$, with $a_j = \mathbb{E}[V \mid C_j]$; it is zero exactly when ACT loses nothing at this composition, and it is estimated by the difference of the validation errors of ACT and MERGE. It is zero whenever both activations are injective functions of their inputs—then (a_1, a_2) determines (C_1, C_2)—whatever the interaction, so it detects the interactions that activations lose, not all interactions.

MERGE and its inverse mirror Cobweb's own merge and split one level up: they decide whether two inputs are perceived by one concept hierarchy or by two.

Procedure E.6 (Task induction by verification)**.**

1. Start with one tree per raw coordinate (or per given block).
2. Estimate the synergy deficit of every pair; merge the pairs whose deficit is significantly positive (union–find over these pairs).
3. Compose the resulting groups bottom-up with two-input trees; at each composition choose the operator in $\{\text{ACT}, \text{LABEL}, \text{HYBRID}, \text{CROSS}\}$ by the validation error of the composing tree's outcome.

By Theorem E.4 each choice in step 3 is consistent given the subtrees below it, among the candidates that satisfy its convergence hypothesis (activation forests under Theorem D.5, label forests under Theorem D.10; hybrid and cross-view compositions are not covered); the procedure as a whole is greedy and carries no global guarantee. By Theorem E.2 no step can be replaced by a rule that looks at a subtree alone. Pairwise deficits are also blind to synergy of higher order: for the parity of three coordinates every pair is independent of Y (Corollary E.3), every pairwise deficit is zero and step 2 merges nothing. The candidates of step 3 must therefore always include the merged tree or a label forest, on which the selection can fall back. Every tree still learns only with `ifit`: the procedure chooses *what* a tree predicts—which observable it will see—never how it learns.

F Measurement details

Trees and prediction. All measurements use the C++ hybrid Cobweb tree of the repository. Constructed trees (Proposition 2 and Theorem 7) are depth-one trees loaded with exactly the statistics of Definition A.1: one leaf per cell centre of an m^d grid on the stated box, holding $(\gamma, g(\gamma))$ with count one, and variance floor $h = \frac{1}{2}\times$ the grid spacing. Learned trees are built by `ifit` alone. Every tree predicts with the count-weighted leaf mixture (1) (the PRIOR weighting restricted to leaves) under full expansion. Learned forests are trained with the split protocol.

R1: one constructed tree (Proposition 2). Targets on $[0,1]^d$: $\sin 2\pi x_1$ $(d=1)$, $\sin(2\pi x_1)\cos(\pi x_2)$ $(d=2)$, $\sin(2\pi x_1)\cos(\pi x_2) + x_3^2$ $(d=3)$; sup error over 400 uniform queries in all of $[0,1]^d$; grids $m = 8, \ldots, 512$ $(d=1)$, $4, \ldots, 64$ $(d=2)$, $3, \ldots, 16$ $(d=3)$.

R2: composition (Theorem 7). The target on $[0,1]^n$, $n = 2, 4, 8$, is the balanced binary composition of $g(a,b) = \frac{1}{2}(a+b) + \frac{1}{4}|\sin(2\pi ab)|$ ("Lipschitz": g has kinks inside every input box, where the Lipschitz rate is attained) or of the same map without the absolute value ("smooth").

target	mixture	$N = 100$	$N = 400$	$N = 1600$	$N = 6400$
$\sin(\pi x)$	leaves only	0.019	0.0066	0.0016	0.00038
$\sin(\pi x)$	path, all	0.043	0.026	0.0074	0.002
$\sin(\pi x)$	lik, all	0.035	0.022	0.0053	0.0012
$\sin(\pi x)$	prior, all	0.197	0.166	0.132	0.112
$\sin(\pi x)$	posterior, all	0.113	0.135	0.085	0.081
$J_0(20x)$	leaves only	0.049	0.015	0.0037	0.0009
$J_0(20x)$	path, all	0.066	0.047	0.016	0.0054
$J_0(20x)$	lik, all	0.059	0.043	0.013	0.0041
$J_0(20x)$	prior, all	0.158	0.150	0.129	0.107
$J_0(20x)$	posterior, all	0.175	0.180	0.183	0.162

Table 2: Interior test RMSE of one learned tree (mean of three seeds), the data of Fig. 3.

Each vertex's input box is the range of its exact inputs over 2×10^4 uniform points, enlarged by 10% on each side (the margin r of Theorem 7). The forest has one constructed two-input tree per vertex with $G = m^2$ leaves, $m = 4, \dots, 64$; the single tree has an m^n grid on $[0,1]^n$. Sup error over 300 uniform queries.

L1: one learned tree (Theorem 8). $Y = g(X) + 0.3\,\xi$ with X uniform on $[0,1]^d$, ξ standard normal and g as in R1; $N \in \{500, \dots, 16000\}$; variance floor $h = \kappa N^{-1/(d+4)}$ in units of each coordinate's standard deviation, $\kappa \in \{0.35, 0.7, 1.4\}$ chosen on 1000 validation points; RMSE to g on 2000 queries in $[0.25, 0.75]^d$.

L2: learned forest and one tree (Proposition 10(a), Remark D.13). $Y = \exp\big(\frac{2}{n}\sum_{j<n/2} \sin \pi(X_{2j}^2 + X_{2j+1}^2)\big) + 0.1\,\xi$, X uniform on $[-1,1]^n$. The forest is the funnel forest over the blocks, composed by a balanced binary graph of two-input trees; each tree's floor follows $h = \kappa N^{-1/(k+4)}$ with k its fan-in. RMSE to f on 2000 queries in $[-0.75, 0.75]^n$, divided by $\mathrm{sd}(f)$.

L3: what trees predict (Proposition 10(b)–(d)). (b),(c): $Y = X_1 X_2 + 0.05\,\xi$ on $[-1,1]^2$; the funnel forest has two one-input trees, the label forest passes labels at a depth chosen in $\{1,2,3,4\}$ on validation data, with smoothing $\alpha = 10^{-4}$ in the output tree (condition (c) of Theorem D.10). (d): the two-view Gaussian model of Proposition D.9 with loadings $(1, 0.8)$ and $(0.9, -0.7)$, noise scales $(0.5, 0.6)$ and $(0.6, 0.5)$, and the view-specific target $Y = E_{1,1}$; the cross-view trees predict one coordinate of the other view. Every variable is linear in a standard Gaussian vector, so $f = \mathbb{E}[Y \mid X]$ and the ideal output $F = \mathbb{E}[Y \mid \mu_1, \mu_2]$ are linear projections and $\Delta/\operatorname{Var} f = \|f - F\|^2/\|f\|^2$ is computed exactly from the coefficient vectors (`experiments/theory_suite.py`, `multiview_deficit`).

G Additional experiments

These experiments use the implementation's defaults (online protocol, PATH rule, a budget of 300 leaves), outside the hypotheses of the convergence theorems; they test the theory's qualitative predictions.

Figure 3 and Table 2 show the four weightings on learned trees. These runs are outside the hypotheses of Theorem B.2 and Proposition B.3, which concern a constructed grid tree, and of Theorem C.2: the variance floor equals the data spacing ($h \asymp 1/N$), the interpolation regime of a noise-free target, not the regime $Nh^d \to \infty$. They show the same pattern as the constructions: for

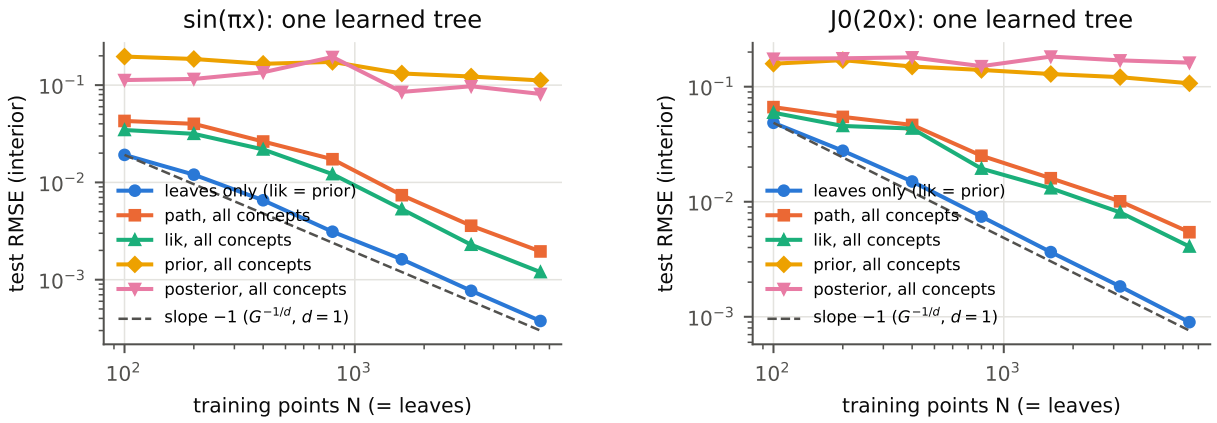


Figure 3: One learned tree, variance floor at the data spacing, full expansion. The leaves-only mixture falls with slope -1; the PATH and LIK mixtures over all concepts fall more slowly; the PRIOR mixture over all concepts falls only like $1/\log N$ and the POSTERIOR mixture stays high (Remark B.4). Numbers in Table 2.

N	leaf budget	κ	level 0 $\lvert S\rvert\le 1$	level 1 $\lvert S\rvert\le 2$	level 2 $\lvert S\rvert\le 4$	level 3 $\lvert S\rvert\le 8$	level 4 $\lvert S\rvert\le 16$	level 5 $\lvert S\rvert\le 32$	level 6 $\lvert S\rvert\le 64$	output RMSE/sd
4000	100	0.25	1.38	1.42	1.40	1.27	1.12	1.08	1.02	1.00
4000	100	1.0	3.16	1.46	1.41	1.38	1.19	1.08	1.06	1.01
4000	1000	1.0	0.57	0.87	1.06	1.10	1.06	1.06	1.02	1.00
4000	4000	1.0	0.57	0.82	0.99	1.02	1.01	1.00	1.00	1.00
16000	1000	1.0	0.90	0.92	1.16	1.07	1.04	0.98	0.97	0.98
16000	4000	1.0	0.56	0.63	0.86	0.91	0.90	0.89	0.89	0.92

Table 3: The 100-D funnel forest level by level (N training points, one seed): RMSE of every tree's activation divided by the standard deviation of its ideal message m_S, averaged over the trees of a level. A ratio near 1 means the activation carries no information about m_S.

$\sin(\pi x)$ the leaves-only error falls from 0.019 at $N = 100$ to 0.00038 at $N = 6400$, a factor of about four per quadrupling of N, while the PRIOR mixture over all concepts stays between 0.11 and 0.20 and the POSTERIOR mixture between 0.08 and 0.14 at the tabulated N (and 0.19 at $N = 800$ in the figure) (between 0.15 and 0.19 for $J_0(20x)$).

Errors compound through levels. Table 3 follows a funnel forest on the 100-dimensional target $\exp\big(\frac{1}{100}\sum_i \sin^2(\pi x_i/2)\big)$ level by level (online protocol, PATH rule, a leaf budget). Each univariate tree must estimate $\mathbb{E}[Y \mid X_i]$, whose standard deviation is about a tenth of that of Y. At $N = 4000$, with a budget of 100 leaves the estimate is dominated by noise (ratio 1.4 for $\kappa = 0.25$, 3.2 for $\kappa = 1$); with 1000 or 4000 leaves it carries signal (ratio 0.57), but the trees one level up read these noisy activations, and the signal is gone by the third level. At $N = 16000$ with 4000 leaves part of it survives to the top (ratios 0.56 to 0.91, output 0.92): with this budget the composition improves with N, in the direction Theorem D.5 predicts for the limit, but with 1000 leaves the low levels get worse with N (0.57 → 0.90 at level 0), so the improvement depends on the budget; these runs are in any case outside the theorem (online protocol, a budget, and messages with critical points, outside (A2)). This is the finite-sample face of the condition behind (A4): the children's errors are

		tree perceiving x_2 alone		composing tree (test RMSE)			
N	world	val. RMSE	act. sd	funnel	labels (d=3)	hybrid	merged
1000	P_a	1.004	0.062	1.008	0.675	0.642	0.270
1000	P_b	1.002	0.080	1.006	1.143	1.116	1.080
4000	P_a	1.000	0.047	1.064	0.460	0.510	0.244
4000	P_b	1.002	0.048	1.028	1.074	1.056	1.018
16000	P_a	1.003	0.069	1.164	0.429	0.471	0.215
16000	P_b	1.004	0.073	1.092	1.051	1.034	1.007

Table 4: The two sign worlds of Theorem E.2 (five seeds). Left: a tree that perceives x_2 and predicts Y—its validation RMSE equals that of the constant predictor (1.00) and its activation is flat in both worlds. Right: the composing tree's test RMSE for four task assignments.

target	N	selected (count of 5)	funnel	selected	best candidate
xy	500	merge (5)	1.020	0.071	0.071
xy	2000	merge (5)	1.000	0.032	0.032
xy	8000	merge (5)	1.003	0.019	0.019
sign	500	merge (5)	0.996	0.309	0.309
sign	2000	merge (5)	1.007	0.245	0.245
sign	8000	merge (5)	1.075	0.231	0.231
expsin	500	merge (5)	0.232	0.115	0.115
expsin	2000	merge (5)	0.145	0.055	0.055
expsin	8000	merge (5)	0.111	0.034	0.034
nested4	500	funnel-pairs (4), funnel (1)	0.773	0.662	0.657
nested4	2000	funnel-pairs (5)	0.651	0.451	0.450
nested4	8000	funnel-pairs (3), hybrid-pairs (2)	0.622	0.334	0.334

Table 5: Task selection by verification (Theorem E.4; five seeds). Candidates: the funnel forest, label forests of depth 2, 3 and 4, a hybrid forest, one merged tree and, for the 4-D target, funnel and hybrid forests over the pairs. Columns: how often each candidate was selected, and the test RMSE/sd of the funnel forest, of the selected forest and of the best candidate in hindsight.

not small compared with what their parents must resolve, and errors compound through the levels.

Locality. Table 4 reproduces Theorem E.2 in data: the tree that perceives x_2 has the same validation error and the same flat activation in both worlds, so nothing it can compute tells it whether x_2 matters. The composing tree separates the worlds: under P_a the merged tree reaches 0.22 and the label forest 0.43 at $N = 16000$, while under P_b, where $f \equiv 0$ and the constant predictor is optimal, every assignment stays near 1. The funnel forest fails under P_a, where its error even rises above the constant predictor's (1.16 at $N = 16000$), because its online, budget-limited activations are noise that the output tree fits.

Selection. Selecting by the validation error at the output (Table 5) never selects the funnel forest on xy or on the sign target. On the 4-D target it selects a forest over the pairs in 14 of 15 runs (in one run at $N = 500$ it selects the univariate funnel forest), over the merged 4-D tree. The selected forest's test error is within one percent of the best candidate's in every cell. On two-dimensional targets the merged tree wins, as it should: it has the same fan-in as the forest and no composition error. These runs test the selection rule rather than instantiate Theorem E.4: the

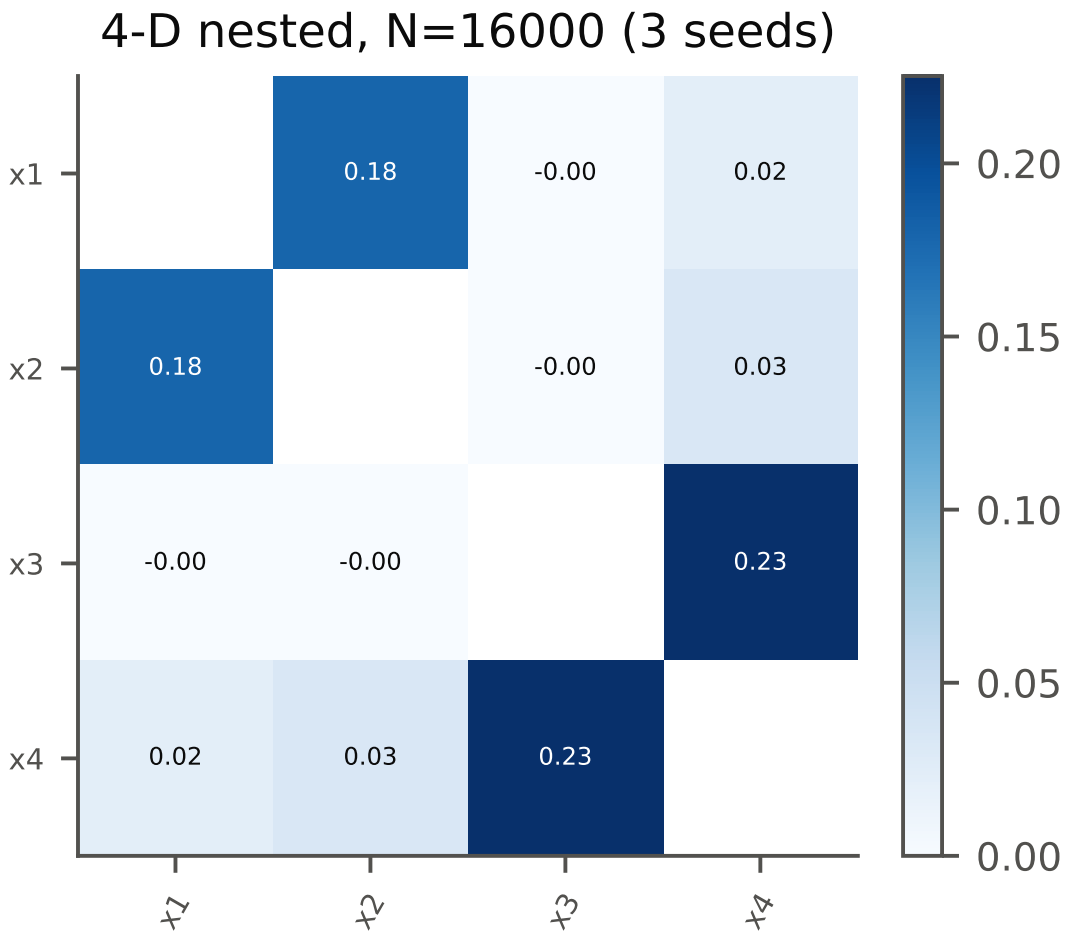


Figure 4: Synergy deficits (Definition E.5): validation MSE of the funnel pair forest minus that of one tree over the pair, divided by $\mathrm{Var}(Y)$. The 4-D target $\exp\big(\frac{1}{2}(\sin\pi(x_1^2+x_2^2)+\sin\pi(x_3^2+x_4^2))\big)$; the two blocks stand out and the cross-block deficits are near zero.

theorem assumes that every candidate converges, whereas here every candidate is trained online with the PATH mixture and a budget, and the fixed-depth label and hybrid candidates are not covered by our consistency results.

Synergy. Figure 4 estimates the synergy deficit of every pair of coordinates, the quantity step 2 of Procedure E.6 thresholds. On the 4-D target the within-block deficits are 0.18 and 0.23 of $\mathrm{Var}(Y)$ and the cross-block deficits are within 0.04 of zero, so the threshold recovers the blocks $\{x_1, x_2\}$ and $\{x_3, x_4\}$. The deficit vanishes whenever both activations are injective functions of their inputs, whatever the interaction (Definition E.5), so a positive deficit marks an interaction that the activations lose, not every interaction.

Task induction. Table 6 runs Procedure E.6 end to end. The groups it forms are the interactions that activations lose: in $q(E_f + Bv\sin\theta)$ it merges B, v and θ and keeps E_f apart, and it also keeps q apart, although q multiplies every term, because q's activation is injective; in $I.6.20b$ it merges θ

target	N	induced groups ($\tau = 0.05$)	one tree	balanced	selected	best	seeds
I.12.11	4000	$\{q\}\{E_f\}\{B, v, \theta\}$	0.216	0.594	0.212	0.212	3
I.13.12	4000	$\{m_1\}\{m_2\}\{r_1\}\{r_2\}\{G\}$	0.311	0.634	0.311	0.311	3
I.29.16	4000	$\{x_1\}\{x_2\}\{\theta_1, \theta_2\}$	0.217	0.853	0.217	0.217	3
I.37.4	4000	$\{I_1\}\{I_2\}\{\delta\}$	0.106	0.299	0.106	0.106	3
I.50.26	4000	$\{x_1\}\{\omega, t\}\{\alpha\}$	0.345	0.826	0.327	0.326	3
I.6.20	4000	$\{\sigma, \theta\}$	0.023	0.350	0.023	0.023	3
I.6.20b	4000	$\{\sigma\}\{\theta, \theta_1\}$	0.083	0.536	0.083	0.083	3
I.9.18	4000	$\{m_1\}\{m_2\}\{x_2\}\{x_1\}\{y_2\}\{y_1\}\{z_2\}\{z_1\}\{G\}$	0.438	0.602	0.438	0.438	3
II.6.15a	4000	$\{\epsilon, r\}\{p_d\}\{x, y\}\{z\}$	0.519	0.562	0.546	0.494	3
nested4	4000	$\{x_1, x_2\}\{x_3, x_4\}$	0.450	0.607	0.407	0.407	3
nested4	16000	$\{x_1, x_2\}\{x_3, x_4\}$	0.300	0.626	0.294	0.294	3

Table 6: Task induction end to end (Procedure E.6 and Procedure 12, steps 1–4; three seeds, $\kappa = 0.25$ for every candidate). Groups formed by step 2 at threshold $\tau = 0.05$ (most frequent over seeds, each run estimating its own deficits) and test RMSE/sd of the single tree, of the generic balanced forest over the inputs, of the forest selected on validation data among the single tree, the balanced forest and the induced forests for $\tau \in \{0.02, 0.05, 0.1, 0.2\}$ with activation or hybrid messages, and of the best candidate in hindsight.

with θ_1, whose difference the formula uses; in $I.29.16$ it merges the two angles; on the 4-D target it recovers the two blocks. The selected forest equals the best candidate, or is within one percent of it, in every row but one; it beats the single tree on the 4-D target and narrowly on $I.12.11$ and $I.50.26$, and it falls back on the single tree on the other formulas. The exception is $II.6.15a$, where the selected forest (0.546) is worse than the single tree (0.519), which was among the candidates. The generic balanced forest is the worst design on every formula: a forest needs the right graph, and verification at the composing tree finds one.

Interpretability. Every tree of a forest can be plotted against its inputs. On $\exp(\sin \pi x_1 + x_2^2)$ the univariate trees learn $c_1 e^{\sin \pi x_1}$ and $c_2 e^{x_2^2}$ (Fig. 5), the inner functions of the target up to the monotone maps H_S of Proposition D.7, without gradients.

Label cells. Theorem D.10 needs the cells of the labels to shrink. In one dimension the cells of a fixed depth shrink slowly as the tree grows; in two dimensions they do not (Fig. 6). Labels become fine enough only if their depth grows with N; the depths our experiments try are fixed (2 to 6), so the label forests of this section are outside Theorem D.10. We also tried the choice of the *basic level*—the concept of largest expected pointwise mutual information on the branch, which Cobweb computes: for continuous attributes it lies near the leaves (mean depth 10 to 11), because a concept's information grows as its variance shrinks towards the variance floor, and its labels did worse than any fixed depth on xy.

No forgetting. Lemma C.1 says that nothing a tree stores is deleted, so learning a new region changes old predictions only through the kernel tails of the new leaves. On the five-peak task (Fig. 7; PATH rule, online) the tree's error on everything seen so far stays between 0.037 and 0.048 through all five phases.

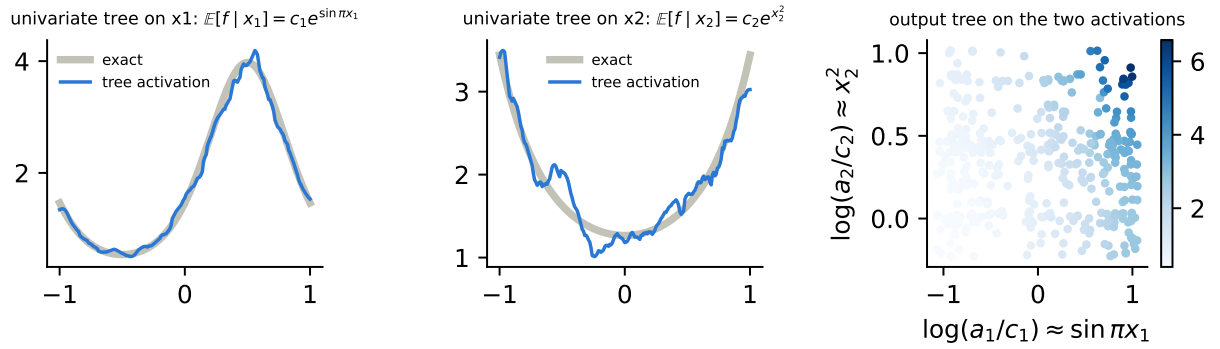


Figure 5: What the univariate trees of a funnel forest learn on $\exp(\sin \pi x_1 + x_2^2)$ ($N = 4000$). Left and centre: each tree's activation against the exact conditional expectation $\mathbb{E}[f \mid x_i]$. Right: the output tree's prediction as a function of the two activations.

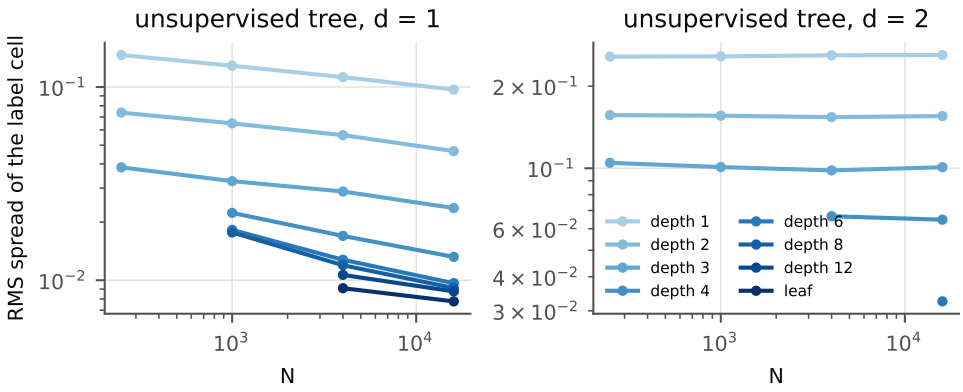


Figure 6: Spread of the cell of a label at fixed depth, for unsupervised trees over uniform data.

H Verification

Executable verifier. `verification/checks.py` instantiates the objects each statement talks about with the real C++ tree—learned by `ifit`, or built exactly for the constructions in the proofs (depth-one grid trees, the label-forest construction, interwoven trees)—and tests the statement's conclusion; `verification/run.py` writes the table below. The first check compares Definition A.3 and Definition A.1, implemented from the definitions, with the code's scores and mixtures.

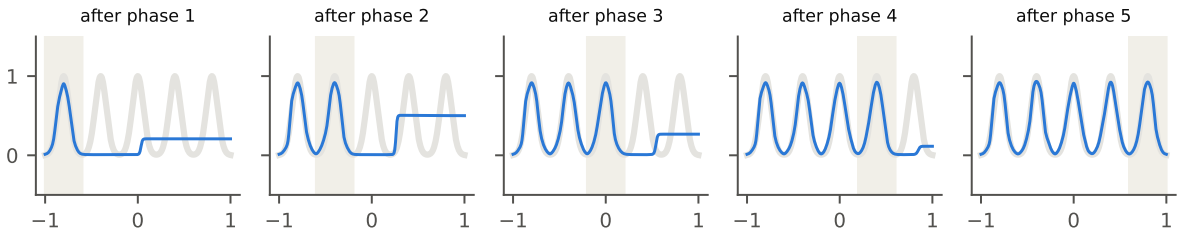


Figure 7: A learned tree on the five-peak continual task of Liu et al. [14, Fig. 3.4]: the data arrive one region at a time (shaded); grey, the target.

result	statement	check
pass	Definition A.3	Def. multi-node prediction and Def. concept equal the implementation
pass	Lemma C.1	Leaf preservation (no leaf deleted; additivity; leaves hold identical copies)
pass	Lemma B.1	Concept-bias certificate holds for every weighting and budget
pass	Theorem B.2, Proposition B.3	Grid tree: error and root weight vanish (lik, path, leaves); prior plateaus
pass	Theorem B.9	Label-forest construction meets its error bound; labels are nearest grid points
pass	Proposition B.11	Interwoven tree = convex combination of the subtrees (lik, path)
pass	Lemma B.5, Theorem B.6	Composed constructed trees approximate $\Phi(\varphi_1 + \varphi_2)$
pass	Theorem C.2	Learned tree, leaves-only lik = Nadaraya–Watson (path is not)
pass	Proposition D.8	xy: funnel activations vanish, funnel forest fails; label error falls with N at every depth
pass	Theorem E.2	Sign construction: local laws equal; composition separates the worlds
pass	Proposition D.9	Posterior given all views = function of per-view posterior means
pass	Remark C.4	Branch-and-bound stop returns a sub-optimal concept (continuous and nominal-only trees)
pass	Proposition D.7	H_S strictly increasing; funnel forest on a monotone-additive target converges
pass	Corollary E.3	3-way parity: pairs independent of Y; label forest learns it, funnel does not
pass	Theorem D.5	Split protocol: sup error to the ideal forest falls with N
pass	Theorem E.4	Validation at the output never selects the funnel forest on xy
pass	Appendix H	Every reported result comes from the fixed build or a path the bug cannot reach
pass	Theorem C.2, Theorem D.10	Count-weighted leaves = Nadaraya–Watson and partitioning estimate with repeats; count-blind leaves fail
pass	Proposition B.3	Binary grid tree: prior (all concepts) keeps falling, posterior (leaves) has a floor, lik (leaves) converges
pass	Appendix H	Lean 4 (Mathlib): every formalised statement builds, with no `sorry` and no added axiom

Formal proofs. The discrete cores of the statements below are proved in Lean 4 with Mathlib [6, 22] (`lean/`, built by `lake build`). The check `check_lean` rebuilds the project, prints the axioms of every listed theorem and fails if any depends on `sorryAx` or if a source file contains `sorry`, `admit` or a new `axiom`; every theorem uses only Lean's standard axioms (`propext`, `Classical.choice`, `Quot.sound`).

statement	Lean theorem	scope
Proposition D.8, Corollary E.3 (constant messages)	`validity_nonneg`, `condExp_const_of_validity_zero`	a tree whose input carries no information about its outcome emits a constant message (any finite distribution)
Theorem E.2(i)–(ii)	`sign_worlds_local_law`, `local_rule_same`, `sign_worlds_differ`	the sign skeleton on $\{\pm 1\}^3$
Corollary E.3	`parity_fibre_balanced`	parity of all n bits ($J = [n]$, signs as bits), every S missing a bit
Lemma C.1 and Theorem C.2 (leaf mixture)	`leaf_sum_eq`, `leaf_mixture_eq_nw`	a count-weighted sum over leaves equals the sum over the examples they hold; hence the leaf mixture is Nadaraya–Watson, for any tree and kernel
Proposition B.11	`merged_prediction`, `merged_weights`	the mixture identity, m subtrees
Proposition B.7	`Chain.fanin_one`	every fan-in-one forest
Theorem B.6 (count)	`ka_tree_count`	$4n^2 + 4n$ trees, $n \geq 1$

Analytic statements (rates, consistency) are not formalised; for them the executable checks and the review below carry the verification.

Adversarial review. Each statement is also checked by an independent reviewer—a fresh language-model agent for every round—instructed to break it, to settle questions by computation, and to read nothing but this document, its tables and the two C++ files it formalises (`verification/REVIEW_PROMPT.md`). The reviewer was calibrated before it was trusted: on a copy of an earlier draft with eight planted errors (three of them real mistakes of that draft) it found all eight with correct diagnoses, and every other flag it raised was a genuine error, which we fixed. The same prompt was then run on successive drafts:

round	outcome
calibration	8/8 planted errors found; extra findings (label consistency with atoms, exact matching, crossover exponents) fixed
2	no statement false under its hypotheses; experiments text contradicting tables, overstated remarks and summaries; fixed
3	no statement false under its hypotheses; prior/posterior weightings on hierarchies (verified by check C19), branch-and-bound repair, minor gaps; fixed
4	no statement invalid; four minor gaps (an unsatisfiable form of a hypothesis of Theorem D.10, an unused density hypothesis, a cited uniform law, an ambiguous phrase) and five text items; fixed
5 (v7, simplified main text)	no statement invalid; every appendix statement valid; main-text summaries had dropped hypotheses (one major: forest convergence needs activation messages and the split protocol), the uniform kernel law became an explicit assumption, a martingale step was added to the prequential remark; fixed

The reports, the drafts they reviewed and the reviewers' own scripts are in `verification/reviews/`; how each finding was resolved is in `verification/REVIEW.md`.

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